# Strict Cubical Horseshoes and All-Tail Full Complexity in Non-Autonomous Dynamics

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**Abstract**

For every cube $Q^d = [0,1]^d$ and every compact smooth $d$-manifold, and for every sequence of dense open windows $W_n$, we construct an open dense set of non-autonomous systems in the full mapping space $C(X,X)^{\mathbb{N}}$ for which every forward tail admits a compact symbolic semifactor onto a prescribed full variable-alphabet shift, a continuum-sized DC1-scrambled set visiting the windows at the prescribed absolute times, infinite topological entropy, and the sharp entropy spectrum $H_{g,r}(\varepsilon) = d\log(1/\varepsilon) + O_g(1)$ with constants independent of the tail $r$. Hence every tail has lower and upper metric mean dimension equal to $d$. The same conclusions hold relatively in the surjective subspaces $\mathcal{F}_s(Q^d)$ and $\mathcal{F}_s(M)$; the surjectivity-preserving implantation proceeds by a disjoint copy core inside each modified region. In particular, the interval theorem and its surjective version are the case $d = 1$.

For the Hilbert cube $Q^\infty = [0,1]^{\mathbb{N}}$ with the standard weighted metric $\rho(x,y) = \sum_j 2^{-j}|x_j - y_j|$, we prove a dense finite-support exact Hilbert-box horseshoe theorem: every tail of every system in the dense class admits a compact symbolic semifactor, a continuum-sized windowed DC1-scrambled set, infinite topological entropy, and for every finite $q \geq 1$ a constant $C_{g,q} > 0$ with $H_{g,r}(\varepsilon) \geq q\log(1/\varepsilon) - C_{g,q}$; consequently both metric mean dimensions are infinite. Proofs are constructive: a strict cubical covering verified by Poincaré–Miranda in finite dimension, copy cores for surjectivity, and exact branch implantation between finite-support Hilbert boxes in the infinite-dimensional case.

**Keywords.** Non-autonomous dynamical systems; Hilbert cube; distributional chaos; DC1 chaos; topological entropy; entropy spectrum; metric mean dimension; Poincaré–Miranda theorem; cubical horseshoes; generic dynamics.

## Contents

# 1 Introduction

A non-autonomous discrete dynamical system on a compact metric space $(X, \rho)$ is a sequence

$$f = (f_1, f_2, f_3, \ldots), \qquad f_n \in C(X, X),$$

whose orbit is

$$x, \quad f_1(x), \quad f_2(f_1(x)), \quad f_3(f_2(f_1(x))), \ldots.$$

Equivalently, one studies the successive compositions

$$f_n \circ f_{n-1} \circ \cdots \circ f_1,$$

rather than the iterates of a single map. The subject therefore combines two kinds of complexity: spatial complexity of the individual maps and temporal complexity of the sequence. It is also naturally suited to tail questions: one may delete any finite initial segment and ask whether the remaining system still displays the same dynamical behavior.

Topological entropy was introduced for autonomous systems by Adler, Konheim and McAndrew [2], with Bowen's separated-set formulation and Walters' exposition providing the standard compact-metric framework [9, 58]. For non-autonomous systems the entropy theory used here goes back to Kolyada and Snoha [32]; interval and piecewise-monotone refinements were developed by Kolyada, Misiurewicz and Snoha [31]. General background on processes, skew-product viewpoints, pullback dynamics and the geometric theory of discrete non-autonomous systems may be found in Kloeden–Rasmussen and Pötzsche [29, 41]. Further aspects of non-autonomous entropy, including metric entropy, variational inequalities, equi-continuity, expansiveness and related structural properties, have been studied for example in [11, 25, 26, 27].

The symbolic side of the paper is classical in spirit. A horseshoe produces a full shift or a full variable-alphabet shift, and symbolic dynamics is one of the standard mechanisms through which complicated compact dynamics are encoded; see Smale's survey, Sinai's Markov partitions, and standard texts [52, 51, 24, 35]. What is non-standard here is not the existence of a symbolic model in one prescribed place, but rather the fact that one can implant a coherent symbolic model in every forward tail of a full non-autonomous mapping space, while also forcing the coded orbit segments to visit externally prescribed dense open windows.

Li–Yorke chaos began with the theorem that period three implies chaos [34]; distributional chaos was introduced by Schweizer and Smítal [45]. The three standard versions DC1, DC2 and DC3 were separated by Balibrea, Smítal and Štefánková [5]; see also the related works of Sklar–Smítal, Oprocha and Downarowicz [50, 40, 16]. In the non-autonomous setting, Li–Yorke chaos, weak mixing, distributional chaos and generic distributional chaos have been studied in many forms, including [10, 59, 7, 17, 53, 46, 4, 6, 3, 8].

**Comparison with Balibrea–Rucká (convergent systems).** The most directly comparable recent result is the Balibrea–Rucká theorem on *convergent* non-autonomous interval systems [8]: in the space of sequences $(f_n)$ converging uniformly to a fixed map, the DC1 systems are dense but *not* open and *not* closed. This is structurally different from the present setting and the apparent tension is only apparent. We work in the full mapping space $C(X,X)^{\mathbb{N}}$ with the uniform-in-time metric $D_X(f,g) = \sup_n d_\infty(f_n, g_n)$, which imposes no convergence hypothesis at all: the perturbed sequence $(g_n)$ need not converge anywhere, and its tails need not converge. Inside this larger space the strict cubical covering relation produced by our construction is open by Poincaré–Miranda margin stability, so the windowed all-tail DC1 class is in fact open dense. Hence the open-density proved here does not contradict the Balibrea–Rucká non-openness phenomenon in the convergent class; it complements it by showing that the obstruction is the convergence constraint, not a property of DC1 itself. The conclusions of this paper are all-tail, windowed and quantitative: symbolic semifactors occur after every finite tail deletion, DC1 scrambled sets visit prescribed dense open windows at prescribed absolute times, entropy is infinite in every tail, and the finite-dimensional entropy spectrum is sharp.

**Relation to previous preprints of the author.** Two earlier preprints by the present author treat the one-dimensional case. The April 2026 preprint [48] establishes *dense* simultaneous distributional chaos and infinite entropy in the full space of non-autonomous interval systems. The May 2026 preprint [49] strengthens this to an *open dense* class via a strict moving-interval horseshoe and obtains generic full small-scale complexity on the interval. The present paper supersedes both interval results: the cubical theorem at $d = 1$ recovers and extends the open dense DC1, infinite entropy, prescribed-window, and full metric mean dimension conclusions of [49], while Corollary 3.4 adds the relative surjective interval theorem that was outside the scope of either preprint. The genuinely new content of the present manuscript, none of which is contained in [48, 49], is:

(a) the $d$-dimensional cubical theorem on $Q^d$ with sharp entropy spectrum $H_{g,r}(\varepsilon) = d\log(1/\varepsilon) + O(1)$ and metric mean dimension $d$ in every tail;

(b) the compact smooth manifold extension, with controlled charts and boundary handling;

(c) the relative surjective theorems on $\mathcal{F}_s(Q^d)$ and $\mathcal{F}_s(M)$ in every finite dimension $d \geq 1$, proved by the copy-core mechanism rather than by a one-dimensional endpoint argument;

(d) all-tail uniformity stated as a single block schedule that meets every prescribed window after every finite tail deletion; and

(e) the Hilbert-cube theorem on $Q^\infty$, with exact finite-support implantation and the infinite-metric-mean-dimension consequence.

In short, [48, 49] together cover the case $d = 1$, and the present paper supplies the finite-dimensional cubical and manifold theory, the relative surjective subspaces, and the infinite-dimensional Hilbert-cube theorem.

Mean dimension was introduced by Gromov [21] and developed by Lindenstrauss and Weiss [38]. Lindenstrauss' embedding work [36], the $\mathbb{Z}^k$-action theory of Gutman–Lindenstrauss–Tsukamoto [22], the full shift computations of Tsukamoto [54], and the rate-distortion variational principle of Lindenstrauss–Tsukamoto [37] form part of the modern background. For non-autonomous systems we use the metric mean dimension formulation of Rodrigues and Acevedo [44]. The generic side is also relevant: Yano proved generic infinite entropy phenomena for autonomous maps in finite-dimensional settings [60]; see also de Faria–Turaev [15] and Hazard's one-dimensional infinite-entropy results [23]. For metric mean dimension, Carvalho–Rodrigues–Varandas proved full upper metric mean dimension for $C^0$-generic homeomorphisms on compact smooth boundaryless manifolds of dimension greater than one [12], while Acevedo proved density and residuality results for prescribed metric mean dimension of continuous maps on compact Riemannian manifolds [1]. The present results differ in that they are non-autonomous, tail-uniform, windowed, and in finite dimension they obtain the sharp entropy spectrum

$$H_{g,r}(\varepsilon) = d\log(1/\varepsilon) + O(1)$$

for every tail, not only the limiting metric-mean-dimension value.

### Horseshoes, covering relations, and windows

The proof mechanism is deliberately topological. In finite dimension, each branch cube is made to stretch across the next cube in every coordinate. The Poincaré–Miranda theorem, a multidimensional intermediate value theorem closely related to Brouwer's fixed point theorem, converts these face inequalities into a covering statement. This places the construction in the line of topological horseshoes, windows and covering relations: Easton's isolating blocks and symbolic dynamics [18], the topological horseshoes and chaos criteria of Kennedy–Yorke and Kennedy–Koçak–Yorke [30, 28], the covering-relation machinery of Zgliczyński and Gidea [19, 20], and later stretching/cutting-surface formulations such as [42, 43, 14]. The Poincaré–Miranda form used in the paper is classical; see Miranda [39], Vrahatis [57], and Kulpa [33].

The word "window" is used here in a slightly different but related sense from the Easton/correctly-aligned-windows tradition. Easton-type windows are usually geometric objects through which orbits are forced to pass in a prescribed order. Our windows are arbitrary dense open subsets $W_n \subset X$ assigned to absolute times. The construction shows that the branch domains can be chosen inside these windows at every time, so that every coded orbit satisfies the external countable list of open requirements. A related dense-open window principle for generic conservative surface semigroups and graph-directed local systems appears in [47]. The present paper proves an analogous principle for full non-autonomous mapping spaces, with entropy-spectrum and metric-mean-dimension control.

### What is new here

The main contribution is the simultaneous combination of the following features.

(i) *Finite cubical core.* The theorem on $Q^d$ is the central finite-dimensional result. It is open dense, tail-uniform, windowed, and sharp at the level of the whole entropy spectrum.

(ii) *All-tail coding.* For every $r \geq 0$, the tail $(g_{r+1}, g_{r+2}, \ldots)$ has its own compact coding set and semiconjugacy onto the same prescribed variable-alphabet scheme, with the first $r$ maps irrelevant.

(iii) *Dense-open windows.* A prescribed sequence of dense open sets $W_n$ is met at the prescribed absolute times by every coded orbit.

(iv) *Sharp finite-dimensional spectrum.* On $Q^d$, and on compact smooth $d$-manifolds, the lower entropy spectrum produced by the horseshoes matches the universal upper covering bound, giving metric mean dimension exactly $d$ in every tail.

(v) *Surjective finite-dimensional subspaces.* The cubical and manifold constructions are made compatible with surjectivity in every finite dimension. The modification of each local cube contains a disjoint copy core on which the new map reproduces the old image of the whole modified cube; hence the image of a surjective map remains the whole space.

(vi) *Hilbert-cube finite-support implantation.* In $Q^\infty$, strict infinite-coordinate covering is not stable in the Hilbert metric. The paper replaces it by exact finite-support Hilbert-box implantation, giving infinite lower and upper metric mean dimension in every tail.

The following table situates the present results in four neighboring areas; it is descriptive and not intended as a ranking.

| Area | Typical setting | Where this paper sits |
|---|---|---|
| Autonomous topological entropy and horseshoes [52, 24, 35, 60, 15, 23] | Symbolic subsystems, generic infinite-entropy phenomena for single maps | Same conclusions transferred to full non-autonomous mapping spaces, uniform over every tail, with prescribed time windows. |
| Non-autonomous entropy and chaos [32, 31, 11, 27, 17, 53, 46, 8] | Entropy theory and DC-type chaos for sequences of maps, often in convergent or otherwise restricted classes | Full uniform-in-time mapping space; open dense finite-dimensional strict schemes, all-tail DC1, and a sharp entropy spectrum. |
| Mean and metric mean dimension [21, 38, 37, 44, 12, 1] | Mean-dimension invariants and generic full metric mean dimension in autonomous settings | Tail-uniform non-autonomous version with a sharp spectrum in finite dimension and infinite metric mean dimension on the Hilbert cube. |
| Covering relations and windows [18, 30, 19, 20, 43] | Topological criteria for symbolic dynamics and transition chains, usually constructed in fixed examples | Dense implantation of such schemes in full sequence spaces, with arbitrary dense-open windows and a finite-support variant on the Hilbert cube. |

## Finite-dimensional result

Let $Q^d = [0,1]^d$ with the sup metric and let

$$\mathcal{F}(Q^d) = C(Q^d, Q^d)^{\mathbb{N}}, \qquad D(f,g) = \sup_{n\geq 1} \sup_{x \in Q^d} \|f_n(x) - g_n(x)\|_\infty.$$

For every sequence of dense open sets $W_n \subset Q^d$, we construct a $D$-open and dense set of systems with strict cubical horseshoe schemes. Each branch cube stretches across the next cube in every coordinate. The Poincaré–Miranda theorem then gives robust covering of the next cube by every branch. A universal block schedule supplies common blocks for proximality, disagreement blocks for DC1 separation, and $m$-branch blocks at separation scale $m^{-1/d}$ for entropy.

The finite-dimensional theorem is sharp. The lower bound

$$H_{g,r}(\varepsilon) \geq d \log(1/\varepsilon) - O_g(1)$$

comes from volume packing at scale $m^{-1/d}$, while the opposite inequality

$$H_{f,r}(\varepsilon) \leq d \log(1/\varepsilon) + O_d(1)$$

holds for every non-autonomous system on $Q^d$, simply by covering orbit segments in $(Q^d)^N$. Thus the constructed systems saturate the whole small-scale entropy spectrum up to an additive constant, and every tail has metric mean dimension exactly $d$. The finite cubical theorem is the main finite-dimensional result of the paper; the manifold theorem and the relative surjective theorems are derived from the same strict covering mechanism with the extra geometric bookkeeping made explicit.

Surjectivity is preserved by a copy-core modification. When a local cube $D$ is changed, a smaller disjoint core $C \subset D$ is mapped through a continuous reparametrization of $D$ onto the old image $f(D)$. Outside the modified cubes the map is unchanged, and on the copy cores the new image contains the old images of the modified cubes. Therefore the new map contains the original image $f(Q^d)$; if $f$ was onto, the perturbation is onto as well. This argument works in every finite dimension and, in controlled charts, on compact smooth manifolds.

### Hilbert-cube result

The Hilbert cube

$$Q^\infty = [0,1]^{\mathbb{N}}, \qquad \rho(x,y) = \sum_{j=1}^{\infty} 2^{-j}|x_j - y_j|$$

is the basic compact model of infinite-dimensional topology; see, for example, Chapman [13] and van Mill [55, 56]. It requires a different horseshoe mechanism. A literal strict cubical covering relation in infinitely many coordinates is not open in the Hilbert metric: high coordinates have arbitrarily small metric weight, so a small $\rho$-perturbation can destroy exact coordinate covering in those directions.

The right replacement is a finite-support Hilbert box: a product box that restricts only finitely many coordinates and leaves the rest free. Such boxes have nonempty interior in $Q^\infty$, so arbitrary dense open windows cannot avoid them. By restricting sufficiently many initial coordinates and using the small tail weight $\sum_{j>K} 2^{-j}$, they can also be made arbitrarily small in $\rho$-diameter. We prove a dense exact implantation theorem: after an arbitrarily small uniform perturbation, each prescribed branch box maps exactly onto the next Hilbert box. This gives all-tail symbolic coding, DC1 chaos, and infinite metric mean dimension.

The Hilbert-cube theorem is stated as a dense theorem, not an open theorem. This is not a weakness of the proof but a genuine feature of exact infinite-coordinate covering in the Hilbert metric. The finite-dimensional strict-margin argument gives open dense sets; the Hilbert-cube finite-support argument gives dense exact systems with infinite complexity in every finite coordinate dimension.

### Organization

Section 2 fixes tail entropy, metric mean dimension and DC1 notation. Section 3 states the main theorems. Sections 4–8 prove the finite cubical theorem, which is the central open-dense result, including the relative surjective subspaces. Section 9 transfers the construction to compact smooth manifolds, including boundary and the relative surjective subspaces. Sections 10–13 prove the Hilbert-cube theorem, and Section 14 records sharpness and scope.

## 2 Tail entropy, metric mean dimension, and DC1

Throughout this section let $(X, \rho)$ be a compact metric space. The space $C(X, X)$ is equipped with the uniform metric

$$d_\infty(u, v) = \sup_{x \in X} \rho(u(x), v(x)),$$

and the full non-autonomous mapping space

$$\mathcal{F}(X) = C(X, X)^{\mathbb{N}}$$

is equipped with the uniform-in-time metric

$$D_X(f, g) = \sup_{n \geq 1} d_\infty(f_n, g_n).$$

This metric is finite because $X$ is compact.

**Lemma 2.1** (Baire property of the uniform-in-time mapping space)**.** *For every compact metric space $X$, the metric space $(\mathcal{F}(X), D_X)$ is complete and hence Baire.*

*Proof.* The space $C(X, X)$ is complete in the uniform metric, since $X$ is compact and the uniform limit of continuous self-maps is continuous. If $(f^{(k)})_{k \geq 1}$ is a $D_X$-Cauchy sequence in $\mathcal{F}(X)$, then for each fixed $n$ the sequence $(f_n^{(k)})_k$ converges uniformly to some $f_n \in C(X, X)$. The Cauchy property is uniform in $n$, so the coordinate limits satisfy $D_X(f^{(k)}, f) \to 0$. Thus $\mathcal{F}(X)$ is complete. The Baire property follows from the Baire category theorem. □

**Lemma 2.2** (Closedness of surjective subspaces)**.** *Let $X$ be a compact metric space, and let*

$$C_s(X, X) = \{u \in C(X, X) : u(X) = X\}.$$

*Then $C_s(X, X)$ is closed in $C(X, X)$. Consequently $\mathcal{F}_s(X) = C_s(X, X)^{\mathbb{N}}$ is closed in $\mathcal{F}(X)$ for the uniform-in-time metric and is a Baire space with the relative metric.*

*Proof.* Let $u_k \in C_s(X, X)$ and suppose $u_k \to u$ uniformly. Fix $y \in X$. For each $k$, choose $x_k \in X$ such that $u_k(x_k) = y$. By compactness, a subsequence $x_{k_j}$ converges to some $x \in X$. Uniform convergence gives

$$u(x) = \lim_{j \to \infty} u_{k_j}(x_{k_j}) = y.$$

Thus $u$ is onto, so $C_s(X, X)$ is closed. If $f^{(k)} \in \mathcal{F}_s(X)$ and $f^{(k)} \to f$ in $D_X$, then for every $n$ the coordinate maps $f_n^{(k)}$ converge uniformly to $f_n$, hence each $f_n$ is surjective. Thus $\mathcal{F}_s(X)$ is closed. Since $\mathcal{F}(X)$ is complete by Lemma 2.1, the closed subspace $\mathcal{F}_s(X)$ is complete and therefore Baire. □

**Definition 2.3** (Tail symbolic semifactor)**.** *Let $f \in \mathcal{F}(X)$. A family of compact sets $K_r \subset X$, finite-alphabet products*

$$\Sigma_r = \prod_{n \geq r+1} A_n,$$

*left shifts $\sigma_r : \Sigma_r \to \Sigma_{r+1}$, and continuous surjections $\Pi_r : K_r \to \Sigma_r$ is called a* tail symbolic semifactor onto the variable-alphabet shift in all tails *if*

$$f_{r+1}(K_r) \subset K_{r+1}, \qquad \Pi_{r+1} \circ f_{r+1} = \sigma_r \circ \Pi_r \quad \text{on } K_r$$

*for every $r \geq 0$. This is a one-sided non-autonomous semiconjugacy convention. Some authors reserve the word "factor" for the case in which the invariant sets are mapped onto one another, or for autonomous systems with a single invariant compact set. Here the natural object is the family of tail coding sets $(K_r)_{r\geq 0}$, and only the inclusion $f_{r+1}(K_r) \subset K_{r+1}$ is needed for the symbolic coding. In the Hilbert-cube exact schemes the individual branch maps are surjective onto the next box, but the global inclusion convention remains the same throughout the paper.*

Let $(X, \rho)$ be compact and let $f = (f_1, f_2, \ldots) \in C(X, X)^{\mathbb{N}}$. For $r \geq 0$, the $r$-th forward tail is

$$\operatorname{tail}^r f = (f_{r+1}, f_{r+2}, \ldots).$$

For $j \geq 1$, put

$$F_r^j = f_{r+j} \circ f_{r+j-1} \circ \cdots \circ f_{r+1}, \qquad F_r^0 = \mathrm{id}_X .$$

The $N$-step Bowen metric of the $r$-th tail is

$$\rho^f_{r,N}(x, y) = \max_{0 \leq j < N} \rho(F_r^j(x), F_r^j(y)).$$

A set $E \subset X$ is $(r, N, \varepsilon)$-separated if

$$\rho^f_{r,N}(x, y) > \varepsilon \qquad (x \neq y,\ x, y \in E).$$

Let

$$s_{r,N}(f, \varepsilon) = \max\{\mathrm{card}(E) : E \subset X \text{ is } (r, N, \varepsilon)\text{-separated}\}.$$

The tail scale entropy is

$$H_{f,r}(\varepsilon) = \limsup_{N\to\infty} \frac{1}{N} \log s_{r,N}(f, \varepsilon).$$

The topological entropy of the $r$-th tail is

$$h(\operatorname{tail}^r f) = \sup_{\varepsilon > 0} H_{f,r}(\varepsilon).$$

The lower and upper metric mean dimensions of the tail, with respect to the fixed metric $\rho$, are

$$\underline{\mathrm{mdim}}_{\mathrm{M}}(\operatorname{tail}^r f) = \liminf_{\varepsilon \downarrow 0} \frac{H_{f,r}(\varepsilon)}{\log(1/\varepsilon)}, \qquad \overline{\mathrm{mdim}}_{\mathrm{M}}(\operatorname{tail}^r f) = \limsup_{\varepsilon \downarrow 0} \frac{H_{f,r}(\varepsilon)}{\log(1/\varepsilon)}.$$

For $x, y \in X$ and $t > 0$, define

$$\psi^r_{xy}(t) = \liminf_{N\to\infty} \frac{1}{N} \#\{0 \leq j < N : \rho(F_r^j(x), F_r^j(y)) < t\}$$

and

$$(\psi^r_{xy})^*(t) = \limsup_{N\to\infty} \frac{1}{N}\#\{0 \le j < N : \rho(F^j_r(x), F^j_r(y)) < t\}.$$

A pair $(x, y)$ is *DC1-scrambled for the $r$-th tail* if

$$(\psi^r_{xy})^*(t) = 1 \quad \text{for every } t > 0,$$

and

$$\psi^r_{xy}(t_0) = 0 \quad \text{for some } t_0 > 0.$$

The scrambled sets below have cardinality $\mathfrak{c} = 2^{\aleph_0}$.

# 3 Main results

## 3.1 Finite-dimensional cubes

Let

$$C_s(Q^d, Q^d) = \{u \in C(Q^d, Q^d) : u(Q^d) = Q^d\}, \qquad \mathcal{F}_s(Q^d) = C_s(Q^d, Q^d)^{\mathbb{N}},$$

with the relative uniform-in-time metric.

**Main Theorem 3.1** (Open-dense full spectrum on cubes; proof at Section 8, p. 22, with the strict cubical scheme constructed across Section 4–Section 7)**.** *Let $d \ge 1$, and let $W_n \subset Q^d$ be dense open sets. There exists a $D$-open and dense set*

$$\mathcal{U}_d(W_\bullet) \subset \mathcal{F}(Q^d)$$

*such that every $g \in \mathcal{U}_d(W_\bullet)$ has the following properties.*

(1) *For every $r \ge 0$, the tail $\operatorname{tail}^r g$ admits a compact coding set giving a tail symbolic semifactor, in the sense of Definition 2.3, onto a full variable-alphabet shift associated with the block schedule of Section 4.*

(2) *For every $r \ge 0$, the tail $\operatorname{tail}^r g$ has a DC1-scrambled set of cardinality $\mathfrak{c}$. The DC1 separation scale may be chosen independently of $r$ and of the pair in the constructed scrambled set.*

(3) *The scrambled set for the $r$-th tail may be chosen so that every coded orbit visits the prescribed windows:*

$$G^j_r(x) \in W_{r+j+1} \qquad (j \ge 0),$$

*where $G^0_r = \mathrm{id}$ and $G^j_r = g_{r+j} \circ \cdots \circ g_{r+1}$.*

(4) *For every $r \ge 0$, $h(\operatorname{tail}^r g) = \infty$.*

(5) *There are constants $C^-_g, C^+_d > 0$, with $C^-_g$ depending on the chosen strict scheme but neither constant depending on $r$, such that for every $r \ge 0$ and all sufficiently small $\varepsilon > 0$,*

$$d \log(1/\varepsilon) - C^-_g \le H_{g,r}(\varepsilon) \le d \log(1/\varepsilon) + C^+_d.$$

(6) *Consequently, for every $r \ge 0$,*

$$\underline{\mathrm{mdim}}_{\mathrm{M}}(\operatorname{tail}^r g) = \overline{\mathrm{mdim}}_{\mathrm{M}}(\operatorname{tail}^r g) = d.$$

*Moreover,*

$$\mathcal{U}_d(W_\bullet) \cap \mathcal{F}_s(Q^d)$$

*is relatively open and dense in $\mathcal{F}_s(Q^d)$, and every system in this relative open dense set has the same all-tail, windowed, symbolic, entropy-spectrum, and metric-mean-dimension conclusions.*

*Where each conclusion is proved.* Openness of $\mathcal{U}_d(W_\bullet)$ is Lemma 6.2. Density (and density relative to $\mathcal{F}_s(Q^d)$) is Lemma 7.1. Conclusion (1) is Proposition 8.2; (2)–(3) are Proposition 8.3; the lower spectrum half of (5) (and (4) as a consequence) is Proposition 8.4; the upper spectrum half of (5) is Lemma 8.5; (6) follows from (5). The surjective-subspace assertion uses Lemma 2.2 together with the surjectivity-preserving statement of Lemma 7.1.

**Corollary 3.2** (Residual finite-dimensional full complexity; proof inline below)**.** *In $\mathcal{F}(Q^d)$, the set of systems such that every forward tail admits a full variable-alphabet shift semifactor, is DC1, has infinite topological entropy, and has lower and upper metric mean dimension $d$, contains a dense open set and therefore is residual. The same statement holds relatively in the surjective subspace $\mathcal{F}_s(Q^d)$.*

**Corollary 3.3** (Countably many window sequences; proof inline below)**.** *Let $\{W_n^a : n \geq 1\}_{a\geq 1}$ be any countable family of dense open window sequences in $Q^d$. There is a residual set $\mathcal{R} \subset \mathcal{F}(Q^d)$ such that, for every $g \in \mathcal{R}$, every $a \geq 1$, and every $r \geq 0$, the $r$-th tail of $g$ has a continuum-sized DC1-scrambled set whose coded orbits visit $W_{r+j+1}^a$ at tail time $j$. The sharp entropy-spectrum and metric-mean-dimension conclusions of Main Theorem 3.1 also hold for every $g \in \mathcal{R}$ and every tail. The same countable-window conclusion holds relatively in $\mathcal{F}_s(Q^d)$.*

*Proof.* Apply Main Theorem 3.1 to each window sequence $W_\bullet^a$. By Lemma 2.1, $(\mathcal{F}(Q^d), D)$ is a Baire space. Hence the intersection $\bigcap_{a\geq 1} \mathcal{U}_d(W_\bullet^a)$ is a dense $G_\delta$, and therefore residual, because every factor is open and dense. The conclusions for the $a$-th window sequence are exactly those of Main Theorem 3.1. Intersecting the relatively open dense sets $\mathcal{U}_d(W_\bullet^a) \cap \mathcal{F}_s(Q^d)$ gives the identical relative statement in $\mathcal{F}_s(Q^d)$, using Lemma 2.2. □

### 3.2 The interval theorem as a corollary

Let $I = [0, 1]$. Since $I = Q^1$, the preceding theorem immediately gives the full interval mapping-space theorem and the relative surjective interval theorem.

**Corollary 3.4** (Interval theorem, including the surjective subspace; proof inline below, as the case $d = 1$ of Main Theorem 3.1)**.** *Let $W_n \subset I$ be dense open sets. There exists a $D$-open dense set*

$$\mathcal{U}_I(W_\bullet) = \mathcal{U}_1(W_\bullet) \subset \mathcal{F}(I)$$

*such that every $g \in \mathcal{U}_I(W_\bullet)$ satisfies all conclusions of Main Theorem 3.1 with $d = 1$. In particular, for every tail $r \geq 0$, $\mathrm{tail}^r g$ has a compact symbolic semifactor, a continuum-sized DC1-scrambled set visiting $W_{r+j+1}$ at tail time $j$, infinite topological entropy, and*

$$H_{g,r}(\varepsilon) = \log(1/\varepsilon) + O_g(1), \qquad \underline{\mathrm{mdim}}_{\mathrm{M}}(\mathrm{tail}^r g) = \overline{\mathrm{mdim}}_{\mathrm{M}}(\mathrm{tail}^r g) = 1.$$

*Moreover, if $\mathcal{F}_s(I) = C_s(I, I)^{\mathbb{N}}$, then*

$$\mathcal{U}_I(W_\bullet) \cap \mathcal{F}_s(I)$$

*is relatively open and dense in $\mathcal{F}_s(I)$, and every system in this relative open dense set has the same all-tail conclusions. Thus the earlier non-tail interval conclusions are recovered by taking $r = 0$.*

*Proof.* This is Main Theorem 3.1 with $d = 1$. The relative statement follows from the surjectivity-preserving part of Lemma 7.1. □

## 3.3 Compact smooth manifolds

For a compact smooth manifold $M$, let

$$C_s(M,M) = \{u \in C(M,M) : u(M) = M\}, \qquad \mathcal{F}_s(M) = C_s(M,M)^{\mathbb{N}},$$

again with the relative uniform-in-time metric.

**Main Theorem 3.5** (Compact smooth manifolds; proof at Section 9, p. 28)**.** *Let $M$ be a compact smooth $d$-dimensional manifold, possibly with boundary, equipped with a fixed Riemannian distance. Assume $d \geq 1$. Let*

$$\mathcal{F}(M) = C(M,M)^{\mathbb{N}}$$

*with the uniform-in-time metric, and let $W_n \subset M$ be dense open sets. There exists a $D$-open and dense set*

$$\mathcal{U}_M(W_\bullet) \subset \mathcal{F}(M)$$

*such that every $g \in \mathcal{U}_M(W_\bullet)$ satisfies, for every tail $r \geq 0$, the following conclusions: the tail admits a full variable-alphabet shift semifactor in the sense of Definition 2.3, has a DC1-scrambled set of cardinality $\mathfrak{c}$ visiting the windows at the prescribed absolute times, has infinite topological entropy, and satisfies*

$$H_{g,r}(\varepsilon) = d\log(1/\varepsilon) + O_g(1) \qquad (\varepsilon \downarrow 0),$$

*where the additive constant may be chosen independently of $r$. Consequently*

$$\underline{\mathrm{mdim}}_{\mathrm{M}}(\mathrm{tail}^r\, g) = \overline{\mathrm{mdim}}_{\mathrm{M}}(\mathrm{tail}^r\, g) = d \qquad (r \geq 0).$$

*Moreover,*

$$\mathcal{U}_M(W_\bullet) \cap \mathcal{F}_s(M)$$

*is relatively open and dense in $\mathcal{F}_s(M)$, and every system in this relative open dense set has the same all-tail, windowed, symbolic, entropy-spectrum, and metric-mean-dimension conclusions.*

*Where each conclusion is proved.* The controlled-chart machinery is Lemma 9.1; openness of $\mathcal{U}_M(W_\bullet)$ and the covering relation are Lemma 9.4; density (and density relative to $\mathcal{F}_s(M)$) is Lemma 9.5. All-tail symbolic, DC1, windowed and entropy lower bounds are Proposition 9.7; the matching upper bound is Lemma 9.6. The surjective-subspace assertion uses the surjectivity-preserving statement of Lemma 9.5.

## 3.4 The Hilbert cube

Let

$$Q^\infty = [0,1]^{\mathbb{N}}, \qquad \rho(x,y) = \sum_{j=1}^{\infty} 2^{-j}|x_j - y_j|,$$

and let

$$\mathcal{F}(Q^\infty) = C(Q^\infty, Q^\infty)^{\mathbb{N}}, \qquad D(f,g) = \sup_{n\geq 1}\sup_{x\in Q^\infty} \rho(f_n(x), g_n(x)).$$

**Main Theorem 3.6** (Dense Hilbert-cube full complexity; proof at Section 13, p. 36, with the exact Hilbert scheme constructed across Section 10–Section 12)**.** *Let $W_n \subset Q^\infty$ be dense open sets. In $\mathcal{F}(Q^\infty)$, the set of systems $g$ satisfying the following properties is dense.*

(1) *For every $r \geq 0$, the tail* $\operatorname{tail}^r g$ *admits a compact coding set giving a tail symbolic semifactor, in the sense of Definition 2.3, onto the full variable-alphabet shift associated with the Hilbert-cube block schedule of Section 12.*

(2) *For every $r \geq 0$, the tail* $\operatorname{tail}^r g$ *has a DC1-scrambled set of cardinality $\mathfrak{c}$, and this set may be chosen so that every coded orbit visits $W_{r+j+1}$ at tail time $j$.*

(3) *For every $r \geq 0$, $h(\operatorname{tail}^r g) = \infty$.*

(4) *For every finite $q \geq 1$ there is a constant $C_{g,q} > 0$, independent of $r$, such that for every $r \geq 0$ and all sufficiently small $\varepsilon > 0$,*

$$H_{g,r}(\varepsilon) \geq q \log(1/\varepsilon) - C_{g,q}.$$

(5) *Consequently, for every $r \geq 0$,*

$$\underline{\mathrm{mdim}}_{\mathrm{M}}(\operatorname{tail}^r g) = \overline{\mathrm{mdim}}_{\mathrm{M}}(\operatorname{tail}^r g) = \infty.$$

*More precisely, each dense system produced below admits a parameter $0 < \alpha < 1/4$ such that for every $r \geq 0$, every finite $q \geq 1$, and every $m \geq 2$,*

$$H_{g,r}\big(2^{-q} c_{\alpha,q} m^{-1/q}\big) \geq \log m,$$

*where $c_{\alpha,q} > 0$ depends only on $\alpha$ and $q$.*

*Where each conclusion is proved.* Finite-support Hilbert boxes and their selection lemmas are Lemmas 10.2 and 10.3. The exact one-step implantation lemma is Lemma 11.3, and the Hilbert universal block schedule is in Section 12. Density of the exact-windowed class is Proposition 12.2. Conclusion (1) is Proposition 13.2; (2) is Proposition 13.3; (3)–(4)–(5), and the more precise lower bound, are Proposition 13.4.

## 4 A finite-dimensional universal block schedule

For the finite-dimensional theorem fix $d \geq 1$. Let

$$\Lambda_d = \{C\} \cup \{D_s : s \geq 1\} \cup \{E_m : m \geq 2\}.$$

The label $C$ will force long common blocks. The label $D_s$ will force long disagreement blocks in the $s$-th binary coordinate of the scrambled-set parameter. The label $E_m$ will force long $m$-branch entropy blocks. Choose a sequence

$$\lambda_1, \lambda_2, \ldots \in \Lambda_d$$

such that every label occurs infinitely often. Choose positive integers $L_k$ so fast that, with

$$S_k = L_1 + \cdots + L_{k-1}, \qquad S_1 = 0,$$

one has

$$\frac{S_k}{L_k} \longrightarrow 0.$$

For instance, after $L_1, \ldots, L_{k-1}$ have been chosen, one may take $L_k > k^2(S_k + 1)$. The $k$-th time block is

$$T_k = \{S_k + 1, S_k + 2, \ldots, S_k + L_k\}.$$

For $n \in T_k$, define

$$M_n = \begin{cases} 2, & \lambda_k = C, \\ 2, & \lambda_k = D_s \text{ for some } s, \\ m, & \lambda_k = E_m. \end{cases}$$

The schedule is tail-universal: for each fixed $r$, along any subsequence of blocks tending to infinity,

$$\frac{S_k - r}{L_k} \longrightarrow 0.$$

# 5 Packing and windowed selection in cubes

**Lemma 5.1** (Separated points in positive-measure sets)**.** *Let $A \subset \mathbb{R}^d$ be bounded and Lebesgue measurable with $\mathrm{Leb}_d(A) > 0$. There is a constant $a_d > 0$, depending only on d, such that for every $m \geq 2$, the set $A$ contains $m$ points whose pairwise sup distances are larger than*

$$a_d \, \mathrm{Leb}_d(A)^{1/d} m^{-1/d}.$$

*One may take $a_d = 1/4$.*

*Proof.* Let $\ell = \mathrm{Leb}_d(A) > 0$, and put

$$r = \frac{1}{16} \left( \frac{\ell}{m} \right)^{1/d}.$$

Suppose that $A$ contains no $m$ points pairwise more than $4r$ apart. Since $A$ is bounded, every $4r$-separated subset of $A$ is finite and there is a maximal one, say $P$. By assumption $\mathrm{card}(P) \leq m - 1$. By maximality, every point of $A$ lies within sup distance $4r$ of some point of $P$; otherwise that point could be added to $P$. Hence $A$ is covered by at most $m - 1$ closed sup-balls of radius $4r$. Each such ball has Euclidean volume $(8r)^d$, and therefore

$$\ell \leq (m-1)(8r)^d < m 8^d 16^{-d} \frac{\ell}{m} = 2^{-d} \ell,$$

a contradiction. Thus $A$ contains $m$ points pairwise more than $4r$ apart, which gives the displayed estimate with $a_d = 1/4$. □

**Lemma 5.2** (Windowed small-image selection in $Q^d$)**.** *Fix $0 < \alpha < 1$ and $d \geq 1$. There is a constant $c_{\alpha,d} > 0$ such that, for every $m \geq 2$, with*

$$\Delta_m(\alpha, d) = c_{\alpha,d} m^{-1/d},$$

*the following holds. Let $J \subset Q^d$ be a closed cube of side length $\alpha$, let $W \subset Q^d$ be dense open, and let $f \in C(Q^d, Q^d)$. Then there are points*

$$p_1, \ldots, p_m \in J \cap W$$

*such that*

$$\|p_i - p_j\|_\infty > 4\Delta_m(\alpha, d) \qquad (i \neq j),$$

*and*

$$\mathrm{diam}_\infty \{f(p_1), \ldots, f(p_m)\} < \alpha/8.$$

*The points may be chosen in the middle subcube of $J$ of side length $\alpha/2$.*

*Proof.* Let $J^*$ be the middle subcube of $J$ of side length $\alpha/2$. Since $Q^d$ is compact, it can be covered by finitely many Borel sets $R_1, \ldots, R_P$, where $P = P_{\alpha,d}$, each of sup diameter at most $\alpha/16$. For example, take a finite grid cover by half-open cubes and assign boundary pieces in any Borel way. The measurable sets

$$A_q = J^* \cap f^{-1}(R_q), \qquad 1 \le q \le P,$$

cover $J^*$. Therefore for some $q$,

$$\mathrm{Leb}_d(A_q) \ge \frac{(\alpha/2)^d}{P}.$$

By Lemma 5.1, $A_q$ contains $m$ points $z_1, \ldots, z_m$ pairwise more than

$$a_d \left( \frac{(\alpha/2)^d}{P} \right)^{1/d} m^{-1/d}$$

apart. Choose

$$c_{\alpha,d} \le \frac{a_d}{8} \left( \frac{(\alpha/2)^d}{P} \right)^{1/d}.$$

Then $\|z_i - z_j\|_\infty > 8\Delta_m(\alpha, d)$ for $i \ne j$. All $f(z_i)$ lie in the same set $R_q$, so their image diameter is at most $\alpha/16$.

By continuity of $f$, choose pairwise disjoint relatively open neighborhoods $U_i \subset J^*$ of $z_i$ such that

$$\mathrm{dist}_\infty(U_i, U_j) > 4\Delta_m(\alpha, d) \quad (i \ne j)$$

and

$$\|f(u) - f(z_i)\|_\infty < \alpha/64 \qquad (u \in U_i).$$

Since $W$ is dense open, each $U_i \cap W$ is nonempty. Choose $p_i \in U_i \cap W$. The separation estimate is immediate, and

$$\mathrm{diam}_\infty\{f(p_1), \ldots, f(p_m)\} \le \alpha/16 + 2(\alpha/64) = 3\alpha/32 < \alpha/8.$$

□

**Remark 5.3** (Choice of separation constants)**.** *The constants $c_{\alpha,d}$ in Lemma 5.2 are not canonical; they may always be decreased. Decreasing them only weakens the required branch separation and leaves the entropy-spectrum lower bounds valid, with a possibly larger additive constant.*

# 6 Strict cubical horseshoe schemes

Let

$$J = \prod_{\ell=1}^{d} [A_\ell, B_\ell] \subset Q^d$$

be a closed cube. If

$$B = \prod_{\ell=1}^{d} [r_\ell, s_\ell]$$

is a branch cube, define its lower and upper $\ell$-faces by

$$F_\ell^-(B) = \{x \in B : x_\ell = r_\ell\}, \qquad F_\ell^+(B) = \{x \in B : x_\ell = s_\ell\}.$$

**Definition 6.1** (Strict windowed cubical horseshoe scheme)**.** *A system $g \in \mathcal{F}(Q^d)$ admits a* strict $W_\bullet$-windowed cubical horseshoe scheme *if there exist $0 < \alpha < 1$, $\theta > 0$, closed cubes*

$$J_n = \prod_{\ell=1}^{d} [A_{n,\ell}, B_{n,\ell}] \subset [2\theta, 1 - 2\theta]^d$$

*of side length $\alpha$, and closed branch cubes*

$$B_{n,1}, \dots, B_{n,M_n} \subset J_n \cap W_n$$

*such that:*

(1) $\operatorname{diam}_\infty(B_{n,i}) < 1/n$.

(2) $\operatorname{dist}_\infty(B_{n,i}, B_{n,j}) > \Delta_{M_n}(\alpha, d)$ *whenever* $i \neq j$.

(3) *For every $n$, $i$, and $\ell$,*

$$\pi_\ell g_n(x) \le A_{n+1,\ell} - 2\theta \qquad (x \in F_\ell^-(B_{n,i})),$$

*and*

$$\pi_\ell g_n(x) \ge B_{n+1,\ell} + 2\theta \qquad (x \in F_\ell^+(B_{n,i})).$$

*Let $\mathcal{U}_d(W_\bullet)$ be the set of all systems admitting such a scheme.*

**Lemma 6.2** (Openness)**.** *For every dense open window sequence $W_n \subset Q^d$, the set $\mathcal{U}_d(W_\bullet)$ is open in $\mathcal{F}(Q^d)$.*

*Proof.* Let $g \in \mathcal{U}_d(W_\bullet)$, and fix a strict scheme with margin $\theta$. Suppose $D(h, g) < \theta$. For $x \in F_\ell^-(B_{n,i})$,

$$\pi_\ell h_n(x) \le \pi_\ell g_n(x) + \theta \le A_{n+1,\ell} - \theta = A_{n+1,\ell} - 2(\theta/2).$$

Similarly, for $x \in F_\ell^+(B_{n,i})$,

$$\pi_\ell h_n(x) \ge B_{n+1,\ell} + \theta = B_{n+1,\ell} + 2(\theta/2).$$

The same cubes and branch cubes still lie in the same windows, and

$$J_n \subset [2\theta, 1 - 2\theta]^d \subset [\theta, 1 - \theta]^d.$$

Thus they define a strict scheme for $h$ with margin $\theta/2$. Hence $B_D(g, \theta) \subset \mathcal{U}_d(W_\bullet)$. □

**Theorem 6.3** (Poincaré–Miranda theorem)**.** *Let*

$$B = \prod_{\ell=1}^{d} [r_\ell, s_\ell] \subset \mathbb{R}^d,$$

*and let $\Phi = (\Phi_1, \dots, \Phi_d) : B \to \mathbb{R}^d$ be continuous. Suppose that for every $\ell$,*

$$\Phi_\ell(x) \le 0 \text{ on } F_\ell^-(B), \qquad \Phi_\ell(x) \ge 0 \text{ on } F_\ell^+(B).$$

*Then there exists $x \in B$ such that $\Phi(x) = 0$.*

**Remark 6.4.** *This is the multidimensional intermediate value theorem of Miranda [39]; it is equivalent to Brouwer's fixed point theorem. See also [57, 33] for short proofs and related forms.*

**Remark 6.5** (Relation with covering relations)**.** *The strict face inequalities in Definition 6.1 are a cubical covering relation in the sense of the topological horseshoe literature. The language used here is intentionally elementary: instead of developing the full covering-relation formalism of [19, 20], we verify the needed covering property directly from Poincaré–Miranda. This choice keeps the perturbation estimates explicit and makes the all-tail non-autonomous bookkeeping transparent. The method is nevertheless close in spirit to Easton's windows [18], Kennedy–Yorke topological horseshoes [30], and the stretching/cutting-surface criteria of [42, 43, 14].*

**Lemma 6.6** (Robust cubical covering)**.** *Let $g$ admit a strict cubical scheme with margin $\theta$. If $D(h,g) < \theta$, then*

$$h_n(B_{n,i}) \supset J_{n+1} \qquad (n \geq 1,\ 1 \leq i \leq M_n).$$

*In particular, $g_n(B_{n,i}) \supset J_{n+1}$.*

*Proof.* Fix $n, i$, and let $y = (y_1, \ldots, y_d) \in J_{n+1}$. Define

$$\Phi_\ell(x) = \pi_\ell h_n(x) - y_\ell, \qquad x \in B_{n,i}.$$

If $x \in F_\ell^-(B_{n,i})$, then by the surviving margin,

$$\Phi_\ell(x) \leq A_{n+1,\ell} - \theta - y_\ell \leq -\theta < 0.$$

If $x \in F_\ell^+(B_{n,i})$, then

$$\Phi_\ell(x) \geq B_{n+1,\ell} + \theta - y_\ell \geq \theta > 0.$$

By Theorem 6.3, there is $x \in B_{n,i}$ with $\Phi(x) = 0$, i.e. $h_n(x) = y$. Since $y \in J_{n+1}$ was arbitrary, $h_n(B_{n,i}) \supset J_{n+1}$. □

# 7 Dense implantation of strict cubical schemes

**Lemma 7.1** (Dense strict implantation, preserving surjectivity)**.** *Let $W_n \subset Q^d$ be dense open sets. For every $f \in \mathcal{F}(Q^d)$ and every $\eta > 0$, there exists $g \in \mathcal{U}_d(W_\bullet)$ such that*

$$D(f,g) < \eta.$$

*If, in addition, $f \in \mathcal{F}_s(Q^d)$, then $g$ may be chosen in $\mathcal{F}_s(Q^d)$. Consequently $\mathcal{U}_d(W_\bullet)$ is dense in $\mathcal{F}(Q^d)$, and $\mathcal{U}_d(W_\bullet) \cap \mathcal{F}_s(Q^d)$ is dense in $\mathcal{F}_s(Q^d)$.*

*Proof.* Choose

$$0 < \alpha < \min\{\eta/4, 1/10\}, \qquad \theta = \alpha/20.$$

Start with any closed cube $J_1 \subset [2\theta, 1-2\theta]^d$ of side length $\alpha$. Suppose $J_n$ has been chosen, and put $m = M_n$. Apply Lemma 5.2 to $J_n$, $W_n$, $f_n$, and $m$. We obtain points $p_{n,1}, \ldots, p_{n,m} \in J_n \cap W_n$ such that

$$\|p_{n,i} - p_{n,j}\|_\infty > 4\Delta_m(\alpha, d) \qquad (i \neq j),$$

and

$$\operatorname{diam}_\infty\{f_n(p_{n,1}), \ldots, f_n(p_{n,m})\} < \alpha/8.$$

By continuity of $f_n$, openness of $W_n$, and the separation of the points, choose pairwise disjoint closed cubes

$$D_{n,1}, \dots, D_{n,m} \subset J_n \cap W_n$$

with $p_{n,i} \in \operatorname{int} D_{n,i}$,

$$\operatorname{dist}_\infty(D_{n,i}, D_{n,j}) > 2\Delta_m(\alpha, d) \qquad (i \neq j),$$

and

$$\operatorname{diam}_\infty f_n(D_{n,1} \cup \dots \cup D_{n,m}) < \alpha/4.$$

Inside each $D_{n,i}$, choose two disjoint closed subcubes

$$B_{n,i},\ C_{n,i} \subset \operatorname{int} D_{n,i}.$$

The cube $B_{n,i}$ is the horseshoe branch; choose it so that

$$\operatorname{diam}_\infty(B_{n,i}) < 1/n, \qquad \operatorname{dist}_\infty(B_{n,i}, B_{n,j}) > \Delta_m(\alpha, d)$$

for $i \neq j$. The cube $C_{n,i}$ is a copy core used only to preserve surjectivity.

Let $c_{n+1} = f_n(p_{n,1})$. Choose

$$J_{n+1} = \prod_{\ell=1}^{d} [A_{n+1,\ell}, B_{n+1,\ell}] \subset [2\theta, 1-2\theta]^d$$

of side length $\alpha$ by setting

$$A_{n+1,\ell} = \min\{\max\{(c_{n+1})_\ell - \alpha/2, 2\theta\}, 1 - 2\theta - \alpha\},$$

and $B_{n+1,\ell} = A_{n+1,\ell} + \alpha$. Because $\alpha + 4\theta < 1$, this is well defined. The enlarged cube

$$J_{n+1}^+ = \prod_{\ell=1}^{d} [A_{n+1,\ell} - 2\theta, B_{n+1,\ell} + 2\theta]$$

is contained in $Q^d$, and every point of $J_{n+1}^+$ is within sup distance at most $\alpha + 4\theta$ of $c_{n+1}$.

We now define $g_n$. Outside $D_{n,1} \cup \dots \cup D_{n,m}$, put $g_n = f_n$. Fix $i$, and write

$$B_{n,i} = \prod_{\ell=1}^{d} [r_\ell, s_\ell].$$

For each $\ell$, define a continuous clipped affine function $q_{i,\ell} : D_{n,i} \to [A_{n+1,\ell} - 2\theta, B_{n+1,\ell} + 2\theta]$ by

$$q_{i,\ell}(x) = \operatorname{clip}_{[A_{n+1,\ell}-2\theta,\, B_{n+1,\ell}+2\theta]}\left(A_{n+1,\ell} - 2\theta + \frac{B_{n+1,\ell} - A_{n+1,\ell} + 4\theta}{s_\ell - r_\ell}(x_\ell - r_\ell)\right).$$

Let $q_i = (q_{i,1}, \dots, q_{i,d})$. Then $q_i(D_{n,i}) \subset J_{n+1}^+$, and on the faces of $B_{n,i}$,

$$\pi_\ell q_i(x) = A_{n+1,\ell} - 2\theta \quad (x \in F_\ell^-(B_{n,i})),$$

while

$$\pi_\ell q_i(x) = B_{n+1,\ell} + 2\theta \quad (x \in F_\ell^+(B_{n,i})).$$

Write

$$D_{n,i} = \prod_{\ell=1}^{d} [u_\ell, v_\ell], \qquad C_{n,i} = \prod_{\ell=1}^{d} [a_\ell, b_\ell].$$

Define a continuous reparametrization $\eta_i : D_{n,i} \to D_{n,i}$ by

$$(\eta_i(x))_\ell = u_\ell + (v_\ell - u_\ell)\,\mathrm{clip}_{[0,1]}\left(\frac{x_\ell - a_\ell}{b_\ell - a_\ell}\right), \qquad 1 \le \ell \le d.$$

Then $\eta_i(C_{n,i}) = D_{n,i}$. Put $r_i = f_n \circ \eta_i$. Choose continuous bump functions $\varphi_i, \psi_i : D_{n,i} \to [0,1]$ such that $\varphi_i = 1$ on $B_{n,i}$, $\psi_i = 1$ on $C_{n,i}$, both vanish on $\partial D_{n,i}$, and $\varphi_i + \psi_i \le 1$. This is possible because $B_{n,i}$ and $C_{n,i}$ are disjoint compact subsets of $\operatorname{int} D_{n,i}$. For $x \in D_{n,i}$, set

$$g_n(x) = (1 - \varphi_i(x) - \psi_i(x)) f_n(x) + \varphi_i(x) q_i(x) + \psi_i(x) r_i(x).$$

Since $Q^d$ is convex, this lies in $Q^d$; since both bump functions vanish on $\partial D_{n,i}$, the pieces glue continuously to $f_n$.

If $x \notin \bigcup_i D_{n,i}$, then $g_n(x) = f_n(x)$. If $x \in D_{n,i}$, then

$$\|f_n(x) - c_{n+1}\|_\infty < \alpha/4, \qquad \|q_i(x) - c_{n+1}\|_\infty \le \alpha + 4\theta.$$

Also $r_i(x) = f_n(\eta_i(x))$, and both $x$ and $\eta_i(x)$ lie in $D_{n,i}$, so

$$\|r_i(x) - f_n(x)\|_\infty < \alpha/4.$$

Therefore

$$\|g_n(x) - f_n(x)\|_\infty \le \max\{\|q_i(x) - f_n(x)\|_\infty, \|r_i(x) - f_n(x)\|_\infty\} < 2\alpha.$$

Thus $\|g_n - f_n\|_\infty < 2\alpha$ for every $n$, and

$$D(f, g) < 2\alpha < \eta.$$

Because $\varphi_i = 1$ on $B_{n,i}$, the strict face inequalities in Definition 6.1 hold. Hence $g \in \mathcal{U}_d(W_\bullet)$.

It remains to record the surjectivity preservation. Assume that $f \in \mathcal{F}_s(Q^d)$. Fix $n$. On $Q^d \setminus \bigcup_i \operatorname{int} D_{n,i}$ the map $g_n$ agrees with $f_n$, and on the copy core $C_{n,i}$ it satisfies

$$g_n(C_{n,i}) = r_i(C_{n,i}) = f_n(\eta_i(C_{n,i})) = f_n(D_{n,i}).$$

Therefore

$$\begin{aligned} g_n(Q^d) &\supset f_n\Big(Q^d \setminus \bigcup_i \operatorname{int} D_{n,i}\Big) \cup \bigcup_i f_n(D_{n,i}) \\ &= f_n(Q^d) = Q^d. \end{aligned}$$

Since $g_n(Q^d) \subset Q^d$, the map $g_n$ is surjective. This holds for every $n$, so $g \in \mathcal{F}_s(Q^d)$. $\square$

# 8 Finite-dimensional coding, DC1, and entropy

Fix $g \in \mathcal{U}_d(W_\bullet)$, and fix a strict scheme. For $r \geq 0$, define

$$G_r^0 = \mathrm{id}_{Q^d}, \qquad G_r^j = g_{r+j} \circ \cdots \circ g_{r+1} \quad (j \geq 1).$$

Let

$$\Sigma_r = \prod_{n \geq r+1} \{1, \dots, M_n\}$$

with the product topology, and let $\sigma_r : \Sigma_r \to \Sigma_{r+1}$ be the left shift.

**Lemma 8.1** (Tail itinerary coding)**.** *Fix $r \geq 0$. For every sequence*

$$\omega = (\omega_{r+1}, \omega_{r+2}, \dots), \qquad \omega_n \in \{1, \dots, M_n\},$$

*there exists $x_\omega \in J_{r+1}$ such that*

$$G_r^j(x_\omega) \in B_{r+j+1, \omega_{r+j+1}} \subset W_{r+j+1} \qquad (j \geq 0).$$

*If two itineraries differ at some time $n$, no point realizes both.*

*Proof.* For a finite itinerary of length $N$, choose any $y_N \in B_{r+N, \omega_{r+N}}$. Since

$$g_{r+N-1}(B_{r+N-1, \omega_{r+N-1}}) \supset J_{r+N}$$

and $y_N \in J_{r+N}$, there exists $y_{N-1} \in B_{r+N-1, \omega_{r+N-1}}$ with $g_{r+N-1}(y_{N-1}) = y_N$. Repeating this backward step gives a point $y_1 \in B_{r+1, \omega_{r+1}} \subset J_{r+1}$ realizing the prescribed finite itinerary.

Thus the compact sets

$$K_N(\omega) = \{x \in J_{r+1} : G_r^j(x) \in B_{r+j+1, \omega_{r+j+1}} \text{ for } 0 \leq j < N\}$$

are nonempty, closed and nested. Their intersection is nonempty, and every point in the intersection realizes the infinite itinerary. If two itineraries differ at time $n$, a common realizing point would put one orbit point in two distinct branch cubes at time $n$, impossible because the branch cubes have positive mutual distance. □

**Proposition 8.2** (Full variable-alphabet shift semifactor)**.** *For every $r \geq 0$, define*

$$\mathcal{K}_r = \{x \in J_{r+1} : G_r^j(x) \in \bigcup_{i=1}^{M_{r+j+1}} B_{r+j+1, i} \textit{ for every } j \geq 0\}.$$

*Then $\mathcal{K}_r$ is a nonempty compact set, $g_{r+1}(\mathcal{K}_r) \subset \mathcal{K}_{r+1}$, and there is a continuous surjection*

$$\Pi_r : \mathcal{K}_r \to \Sigma_r$$

*such that*

$$\Pi_{r+1} \circ g_{r+1} = \sigma_r \circ \Pi_r \qquad \textit{on } \mathcal{K}_r.$$

*Proof.* The set $\mathcal{K}_r$ is an intersection of closed subsets of the compact cube $J_{r+1}$, hence compact. It is nonempty by Lemma 8.1. If $x \in \mathcal{K}_r$, then the forward orbit of $g_{r+1}(x)$ under the $(r+1)$-st tail satisfies the same branch condition with the first symbol removed, so $g_{r+1}(x) \in \mathcal{K}_{r+1}$.

For $x \in \mathcal{K}_r$, the branch cubes at each time are pairwise disjoint; hence there is a unique itinerary $\Pi_r(x) \in \Sigma_r$ satisfying

$$G_r^j(x) \in B_{r+j+1,(\Pi_r(x))_{r+j+1}} \qquad (j \geq 0).$$

Surjectivity follows from Lemma 8.1. To prove continuity, fix $n \geq r+1$. The compact branch cubes $B_{n,i}$ have positive mutual distance. Therefore, relative to $\mathcal{K}_r$, membership of $G_r^{n-r-1}(x)$ in a specified $B_{n,i}$ is locally constant. Thus every finite cylinder in $\Sigma_r$ has open preimage, and $\Pi_r$ is continuous. The identity with the shift follows by deleting the first branch symbol. $\square$

**Proposition 8.3** (Continuum-sized windowed tail DC1)**.** *For every $r \geq 0$, the tail* $\mathrm{tail}^r g$ *has a DC1-scrambled set $\mathcal{S}_r$ of cardinality $\mathfrak{c}$, and*

$$G_r^j(x) \in W_{r+j+1} \qquad (x \in \mathcal{S}_r,\ j \geq 0).$$

*The same separation scale $\Delta_2(\alpha, d)$ works for every tail and every distinct pair in the constructed scrambled sets.*

*Proof.* For each binary sequence $\beta = (\beta_1, \beta_2, \ldots) \in \{0,1\}^{\mathbb{N}}$, define an itinerary $\omega^\beta$ for times $n \geq r+1$. On a $C$-block put $\omega_n^\beta = 1$. On a $D_s$-block put $\omega_n^\beta = 1 + \beta_s$. On every $E_m$-block put $\omega_n^\beta = 1$. Choose a realizing point $x_\beta^{(r)}$ from Lemma 8.1, and set

$$\mathcal{S}_r = \{x_\beta^{(r)} : \beta \in \{0,1\}^{\mathbb{N}}\}.$$

The window condition follows from Lemma 8.1. If $\beta \neq \gamma$, choose $s$ such that $\beta_s \neq \gamma_s$. Since $D_s$ occurs infinitely often, there is a $D_s$-block after time $r$, and the itineraries differ on that block. By Lemma 8.1, no point realizes both itineraries; hence $x_\beta^{(r)} \neq x_\gamma^{(r)}$. Thus $\mathrm{card}(\mathcal{S}_r) = \mathfrak{c}$.

Let $x = x_\beta^{(r)}$ and $y = x_\gamma^{(r)}$ be distinct. Fix $t > 0$. Choose $N_t$ so large that $1/n < t$ for all $n \geq N_t$. Along every $C$-block with $S_k + 1 \geq \max\{r+1, N_t\}$, both itineraries choose branch 1, so at all times $n \in T_k$ the two orbit points lie in the same branch cube $B_{n,1}$, whose diameter is less than $1/n < t$. With $N_k = S_k + L_k - r$, the fraction of $t$-close times among $0, \ldots, N_k - 1$ is at least

$$\frac{L_k}{S_k + L_k - r}.$$

Along the infinitely many $C$-blocks this tends to 1, because $(S_k - r)/L_k \to 0$. Therefore $(\psi_{xy}^r)^*(t) = 1$. Since $t > 0$ was arbitrary, the upper distribution function is identically 1 on $(0, \infty)$.

Choose $s$ with $\beta_s \neq \gamma_s$, and consider $D_s$-blocks after $r$. On such a block the two itineraries choose branches 1 and 2. These branches have mutual distance larger than $t_0 = \Delta_2(\alpha, d)$. Hence throughout the block the two tail orbits are not $t_0$-close. Therefore, along the endpoints of those blocks,

$$\frac{1}{S_k + L_k - r} \#\{0 \leq j < S_k + L_k - r : \|G_r^j(x) - G_r^j(y)\|_\infty < t_0\} \leq \frac{S_k - r}{S_k + L_k - r} \longrightarrow 0.$$

Thus $\psi_{xy}^r(t_0) = 0$, and the pair is DC1-scrambled. $\square$

**Proposition 8.4** (Lower entropy spectrum on cubes)**.** *Let $g \in \mathcal{U}_d(W_\bullet)$, and fix a strict scheme with scale $\alpha$. Then, for every tail $r \geq 0$ and every $m \geq 2$,*

$$H_{g,r}(\Delta_m(\alpha, d)) \geq \log m.$$

*Consequently, there is a constant $C_g^-$, independent of $r$, such that for every $r \geq 0$ and all sufficiently small $\varepsilon > 0$,*

$$H_{g,r}(\varepsilon) \geq d \log(1/\varepsilon) - C_g^-.$$

*Proof.* Fix $r \geq 0$ and $m \geq 2$. Since $E_m$ occurs infinitely often, choose $E_m$-blocks

$$T_k = \{S_k + 1, \dots, S_k + L_k\}$$

with $S_k + 1 \geq r + 1$. For each word

$$u = (u_{S_k+1}, \dots, u_{S_k+L_k}) \in \{1, \dots, m\}^{L_k},$$

use the finite part of Lemma 8.1 to choose a point $x_u \in J_{r+1}$ whose itinerary uses the symbols $u_n$ on $T_k$ and symbol 1 outside $T_k$ before the end of the block. If $u \neq v$, then $u_n \neq v_n$ for some $n \in T_k$. At tail time $j = n - r - 1$, the orbit points lie in different branch cubes $B_{n,u_n}$ and $B_{n,v_n}$, whose mutual distance is larger than $\Delta_m(\alpha, d)$. Therefore $\{x_u : u \in \{1, \dots, m\}^{L_k}\}$ is $(r, S_k + L_k - r, \Delta_m(\alpha, d))$-separated and has cardinality $m^{L_k}$. Taking logarithms, dividing by $S_k + L_k - r$, and passing to the limsup along $E_m$-blocks gives

$$H_{g,r}(\Delta_m(\alpha, d)) \geq \log m.$$

Since $\Delta_m(\alpha, d) = c_{\alpha,d} m^{-1/d}$, choose for each sufficiently small $\varepsilon > 0$ an integer $m \geq 2$ such that

$$c_{\alpha,d}(m+1)^{-1/d} < \varepsilon \leq c_{\alpha,d} m^{-1/d}.$$

The monotonicity of $H_{g,r}$ in the scale gives

$$H_{g,r}(\varepsilon) \geq H_{g,r}(c_{\alpha,d} m^{-1/d}) \geq \log m.$$

The choice of $m$ implies

$$m \geq (c_{\alpha,d}/\varepsilon)^d - 1.$$

For sufficiently small $\varepsilon$, this yields

$$\log m \geq d \log(1/\varepsilon) - C_g^-,$$

where $C_g^-$ depends only on $\alpha$ and $d$, and hence not on $r$. $\square$

**Lemma 8.5** (Universal cubical upper entropy spectrum)**.** *For every $f \in \mathcal{F}(Q^d)$ and every $r \geq 0$, there is a constant $C_d^+ > 0$ such that, for all sufficiently small $\varepsilon > 0$,*

$$H_{f,r}(\varepsilon) \leq d \log(1/\varepsilon) + C_d^+.$$

*Proof.* Cover $Q^d$ by at most $C_d \varepsilon^{-d}$ sets of sup diameter at most $\varepsilon$. For $N \geq 1$, consider the orbit map

$$\Gamma_{r,N} : Q^d \to (Q^d)^N, \qquad \Gamma_{r,N}(x) = (G_r^0(x), G_r^1(x), \dots, G_r^{N-1}(x)).$$

Equip $(Q^d)^N$ with the sup product metric. If $E$ is $(r, N, \varepsilon)$-separated, then $\Gamma_{r,N}(E)$ is $\varepsilon$-separated. The product cover has at most $(C_d \varepsilon^{-d})^N$ sets of product diameter at most $\varepsilon$, and each contains at most one point of $\Gamma_{r,N}(E)$. Hence

$$s_{r,N}(f, \varepsilon) \leq (C_d \varepsilon^{-d})^N.$$

Taking logarithms, dividing by $N$, and taking lim sup proves the claim with $C_d^+ = \log C_d$. $\square$

*Proof of Main Theorem 3.1.* By Lemma 6.2, $\mathcal{U}_d(W_\bullet)$ is open; by Lemma 7.1, it is dense. The relative openness of $\mathcal{U}_d(W_\bullet) \cap \mathcal{F}_s(Q^d)$ in $\mathcal{F}_s(Q^d)$ follows from the openness in the full space, and the relative density follows from the surjectivity-preserving part of Lemma 7.1.

For every $g \in \mathcal{U}_d(W_\bullet)$ and every $r \geq 0$, Proposition 8.2 gives the full shift semifactor, and Proposition 8.3 gives windowed DC1 chaos. The lower spectrum estimate is Proposition 8.4, and the upper spectrum estimate is Lemma 8.5. The lower estimate implies infinite entropy, while

$$d - \frac{C_g^-}{\log(1/\varepsilon)} \leq \frac{H_{g,r}(\varepsilon)}{\log(1/\varepsilon)} \leq d + \frac{C_d^+}{\log(1/\varepsilon)}$$

for small $\varepsilon$. Letting $\varepsilon \downarrow 0$ gives both lower and upper metric mean dimensions equal to $d$, uniformly for every tail. These conclusions apply in particular to the relative surjective open dense subset. $\square$

*Proof of Corollary 3.2.* Take $W_n = Q^d$ for every $n$. The desired class contains the dense open set $\mathcal{U}_d((Q^d)_\bullet)$, hence is residual. Its intersection with $\mathcal{F}_s(Q^d)$ contains the relatively dense open set $\mathcal{U}_d((Q^d)_\bullet) \cap \mathcal{F}_s(Q^d)$, hence is residual in $\mathcal{F}_s(Q^d)$ by Lemma 2.2. $\square$

# 9 Compact smooth manifolds

We prove Main Theorem 3.5 by carrying the cubical construction in uniformly controlled charts. Boundary charts are allowed; their coordinate images lie in the half-space

$$\mathbb{H}^d = \{u = (u_1, \ldots, u_d) \in \mathbb{R}^d : u_d \geq 0\}.$$

**Lemma 9.1** (Uniform controlled coordinate boxes)**.** *There exist constants $\alpha_0 > 0$, $L \geq 1$, $A \geq 1$, $B \geq 1$, a finite family of coordinate charts $(U_a, \chi_a)$, and a chart margin function $\mu_{\text{atlas}} : (0, \alpha_0) \to (0, \infty)$ depending only on $M$ and the chosen finite atlas, with the following property. For every $0 < \alpha < \alpha_0$ and every $y \in M$, there is a chart $(U_a, \chi_a)$ and a closed coordinate cube*

$$J = \chi_a^{-1}\Big(\prod_{\ell=1}^{d}[A_\ell, A_\ell + \alpha]\Big) \subset U_a \cap \operatorname{int} M$$

*such that:*

(i) $\rho(y, J) \leq A\alpha$.

(ii) *The Riemannian $B\alpha$-neighborhood of $\{y\} \cup J$, and the coordinate $8\alpha$-enlargement of $J$, are contained in $U_a$. On this neighborhood,*

$$L^{-1}\|\chi_a(x) - \chi_a(z)\|_\infty \leq \rho(x, z) \leq L\|\chi_a(x) - \chi_a(z)\|_\infty.$$

(iii) *The coordinate convex hull of $\chi_a(y)$ and the coordinate $8\alpha$-enlargement of $\chi_a(J)$ is contained in $\chi_a(U_a)$. In a boundary chart this convex hull lies in $\mathbb{H}^d$, with normal coordinate bounded below by $2\alpha$; the bound is preserved under any further convex combination because $\mathbb{H}^d$ is convex.*

(iv) *For every $0 < \alpha < \alpha_0$ and every $y \in M$, the coordinate $4\alpha$-enlargement of $\chi_a(J)$ has coordinate $\ell_\infty$-distance at least $\mu_{\text{atlas}}(\alpha) > 0$ from $\mathbb{R}^d \setminus \chi_a(U_a)$. The function $\mu_{\text{atlas}}$ is decreasing in $\alpha$, satisfies the explicit lower bound $\mu_{\text{atlas}}(\alpha) \geq r_0 - 4\alpha$ where $r_0 > 0$ is the uniform coordinate radius below, and is independent of $y$ and of the absolute time index at which the box is later used.*

*Proof.* Choose a finite smooth atlas $\{(V_a, \chi_a)\}$, using boundary charts into $\mathbb{H}^d$ near $\partial M$, and choose smaller relatively compact domains $V_a' \Subset V_a$ still covering $M$. After shrinking once more, there is a uniform coordinate radius $r_0 > 0$ such that for every $y \in V_a'$ the coordinate sup ball of radius $r_0$ around $\chi_a(y)$, intersected with $\mathbb{H}^d$ in boundary charts, is contained in $\chi_a(V_a)$. Since the metric coefficients are continuous on finitely many compact chart closures, there is a uniform $L \geq 1$ for which the Riemannian metric and the coordinate sup norm are $L$-bi-Lipschitz on these coordinate balls.

Take $\alpha_0 > 0$ so small that $30\alpha_0 < r_0$. Let $0 < \alpha < \alpha_0$ and $y \in M$, and choose a chart with $y \in V_a'$. In an interior chart, place a coordinate cube of side $\alpha$ within coordinate distance $2\alpha$ of $\chi_a(y)$. In a boundary chart, write $\chi_a(y) = (v, h)$, with $h \geq 0$, and place the cube with its tangential coordinates within $2\alpha$ of $v$ and with lower normal coordinate $h + 10\alpha$. Then the cube lies in the interior of the half-space, and its coordinate $8\alpha$-enlargement has normal coordinate at least $h + 2\alpha \geq 2\alpha$. Hence the enlargement, and also the coordinate convex hull with $\chi_a(y)$, remain in $\mathbb{H}^d$ with normal coordinate $\geq 2\alpha$; any further convex combination of points satisfying this bound also lies in $\mathbb{H}^d$ by convexity. In both the interior and boundary cases, the cube and its $8\alpha$-enlargement lie inside the coordinate ball of radius $30\alpha < r_0$, so they are contained in $U_a$. The Riemannian-distance and bi-Lipschitz estimates follow from the uniform chart bounds, after increasing the constants $A$ and $B$ if necessary.

For property (iv), put

$$\mu_{\text{atlas}}(\alpha) = r_0 - 4\alpha.$$

This is strictly positive for $0 < \alpha < \alpha_0$ (since $30\alpha_0 < r_0$), decreasing in $\alpha$, and depends only on $M$ and the finite atlas. The coordinate $4\alpha$-enlargement of $\chi_a(J)$ sits inside the coordinate ball of radius $r_0 - \mu_{\text{atlas}}(\alpha) = 4\alpha$ centered at one of its corners, so its coordinate sup distance from $\mathbb{R}^d \setminus \chi_a(U_a)$ is at least $\mu_{\text{atlas}}(\alpha)$. The construction never depended on the choice of $y$ other than to select the chart index $a$, so this margin is uniform in $y$ and in the absolute time index $n$ at which the box is later used. ☐

**Lemma 9.2** (Windowed small-image chart selection)**.** *Fix $0 < \alpha < \alpha_0$. There is a constant $c_{\alpha,M} > 0$ such that, for every $m \geq 2$, with*

$$\Delta_m^M(\alpha) = c_{\alpha,M} m^{-1/d},$$

*the following holds. Let $J \subset M$ be a closed coordinate cube of coordinate side length $\alpha$ inside one of the controlled charts, let $W \subset M$ be dense open, and let $f \in C(M, M)$. Then there are points $p_1, \ldots, p_m \in J \cap W$ whose mutual Riemannian distances are larger than $4\Delta_m^M(\alpha)$, and whose images under $f$ have Riemannian diameter less than $\alpha$. The points may be chosen in the coordinate middle subcube of $J$.*

*Proof.* Work in the source chart $\chi$ containing $J$, and let $J^*$ be the coordinate middle subcube of side $\alpha/2$. Cover compact $M$ by finitely many Borel sets $R_1, \ldots, R_P$, each of Riemannian diameter less than $\alpha/4$. The measurable sets

$$A_q = \chi(J^* \cap f^{-1}(R_q)), \qquad 1 \leq q \leq P,$$

cover the coordinate cube $\chi(J^*)$. Therefore one has coordinate Lebesgue measure at least $(\alpha/2)^d/P$. Applying Lemma 5.1 in coordinates gives $m$ points in that cell with coordinate sup separation at least a constant times $m^{-1/d}$. Using the uniform lower Lipschitz bound from Lemma 9.1, decrease the constant so that the Riemannian separation is larger than $8\Delta_m^M(\alpha)$.

All selected images lie in one $R_q$. By continuity, choose small pairwise disjoint relative coordinate neighborhoods of the selected points inside $J^*$, preserving Riemannian separation larger than $4\Delta_m^M(\alpha)$ and moving images by less than $\alpha/8$. Each neighborhood meets $W$, because $W$ is dense open. Choosing one point in each intersection gives image diameter less than $\alpha/4 + 2(\alpha/8) < \alpha$ and the required separation. $\square$

A coordinate cube $J$ has coordinate faces defined by the faces of $\chi(J)$. If

$$\chi(J) = \prod_{\ell=1}^{d} [A_\ell, B_\ell],$$

write $J^+(\theta)$ for the coordinate enlargement

$$\chi^{-1}\left(\prod_{\ell=1}^{d} [A_\ell - 2\theta, B_\ell + 2\theta]\right),$$

whenever it is contained in the chart domain.

**Definition 9.3** (Strict manifold horseshoe scheme)**.** *A system $g \in \mathcal{F}(M)$ admits a strict $W_\bullet$-windowed manifold horseshoe scheme if there exist $0 < \alpha < \alpha_0$, $\theta > 0$, controlled coordinate cubes $J_n \subset M$ of coordinate side length $\alpha$, and branch coordinate cubes*

$$B_{n,1}, \ldots, B_{n,M_n} \subset J_n \cap W_n$$

*such that:*

(1) $\operatorname{diam}_\rho(B_{n,i}) < 1/n$.

(2) $\operatorname{dist}_\rho(B_{n,i}, B_{n,j}) > \Delta_{M_n}^M(\alpha)$ *for* $i \neq j$.

(3) *If $\xi_n$ is the source chart for $J_n$ and $\zeta_{n+1}$ is the target chart for $J_{n+1}$, with*

$$\zeta_{n+1}(J_{n+1}) = \prod_{\ell=1}^{d} [A_{n+1,\ell}, B_{n+1,\ell}],$$

*then $J_{n+1}^+(\theta)$ is contained in the target chart. In addition, there is $\mu > 0$, independent of $n$ and $i$, such that the $\rho$-neighborhood of radius $\mu$ around $g_n(B_{n,i})$ is contained in the same target chart for every $n, i$.*

(4) *For every $n, i, \ell$,*

$$\pi_\ell(\zeta_{n+1} \circ g_n)(x) \leq A_{n+1,\ell} - 2\theta \quad (x \in F_\ell^-(B_{n,i}) \textit{ in source coordinates}),$$

*and*

$$\pi_\ell(\zeta_{n+1} \circ g_n)(x) \geq B_{n+1,\ell} + 2\theta \quad (x \in F_\ell^+(B_{n,i}) \textit{ in source coordinates}).$$

*Let $\mathcal{U}_M(W_\bullet)$ be the set of systems admitting such a scheme.*

**Lemma 9.4** (Manifold openness and covering)**.** *The set $\mathcal{U}_M(W_\bullet)$ is open. Moreover, if $h$ is sufficiently $D$-close to $g \in \mathcal{U}_M(W_\bullet)$, then*

$$h_n(B_{n,i}) \supset J_{n+1} \qquad (n \geq 1,\ 1 \leq i \leq M_n).$$

*Proof.* Fix a strict manifold scheme for $g$, with coordinate margin $\theta$ and chart margin $\mu$. Because only finitely many controlled chart types are used and the relevant sets have a uniform positive chart margin, there exists $\delta > 0$, with $\delta < \mu/2$, such that whenever $\rho(y, z) < \delta$ and $y \in g_n(B_{n,i})$, the point $z$ remains in the same target chart and

$$\|\zeta_{n+1}(y) - \zeta_{n+1}(z)\|_\infty < \theta.$$

If $D(h, g) < \delta$, then the target-coordinate face inequalities survive with margin $\theta/2$, and the images $h_n(B_{n,i})$ remain in the same charts. The same coordinate cubes therefore define a strict scheme for $h$, with smaller margin. This proves openness.

For covering, fix $y \in J_{n+1}$. In target coordinates define

$$\Phi_\ell(x) = \pi_\ell \zeta_{n+1}(h_n(x)) - \pi_\ell \zeta_{n+1}(y)$$

on the source coordinate cube $B_{n,i}$. The surviving face inequalities put $\Phi_\ell \le 0$ on the lower $\ell$-face and $\Phi_\ell \ge 0$ on the upper $\ell$-face. By Poincaré–Miranda there is $x \in B_{n,i}$ with $\Phi(x) = 0$, hence $h_n(x) = y$. Thus $h_n(B_{n,i}) \supset J_{n+1}$. □

**Lemma 9.5** (Manifold dense implantation, preserving surjectivity)**.** *For every $f \in \mathcal{F}(M)$ and every $\eta > 0$, there exists $g \in \mathcal{U}_M(W_\bullet)$ with $D(f, g) < \eta$. If $f \in \mathcal{F}_s(M)$, then $g$ may be chosen in $\mathcal{F}_s(M)$.*

*Proof.* Choose $0 < \alpha < \alpha_0$ so small that every coordinate displacement of sup-size at most $C^*\alpha$ has Riemannian size less than $\eta$, where $C^* = L(B + 12)$ is the explicit controlled constant arising in the estimate at the end of the proof; here $L, B$ are from Lemma 9.1 and are independent of $n$ and $y$. Put $\theta = \alpha/20$, and let

$$\mu_* = \mu_{\text{atlas}}(\alpha) > 0$$

be the chart margin supplied by Lemma 9.1(iv); note that $\mu_*$ is the same at every absolute time $n$ below. Start from any controlled coordinate cube $J_1$ of coordinate side length $\alpha$.

Assume $J_n$ has been chosen, and put $m = M_n$. Apply Lemma 9.2 to obtain well-separated points

$$p_{n,1}, \ldots, p_{n,m} \in J_n \cap W_n$$

whose images under $f_n$ have Riemannian diameter less than $\alpha$. By continuity, openness of $W_n$, and separation, choose pairwise disjoint closed coordinate cubes

$$D_{n,1}, \ldots, D_{n,m} \subset J_n \cap W_n$$

around these points. Inside each $D_{n,i}$, choose two disjoint smaller closed coordinate cubes

$$B_{n,i},\ C_{n,i} \subset \operatorname{int} D_{n,i}.$$

The cube $B_{n,i}$ is the horseshoe branch; choose it so that the branch separation is greater than $\Delta_m^M(\alpha)$ and the branch diameters are $< 1/n$. The cube $C_{n,i}$ is a copy core. Shrinking the $D_{n,i}$'s if necessary, while preserving all these properties, arrange

$$\operatorname{diam}_\rho f_n(D_{n,1} \cup \cdots \cup D_{n,m}) < 2\alpha.$$

Let $c_{n+1} = f_n(p_{n,1})$. By Lemma 9.1, choose a target coordinate cube $J_{n+1}$ of side $\alpha$, with target chart $\zeta_{n+1}$, such that $\rho(c_{n+1}, J_{n+1}) \le A\alpha$, the coordinate $8\alpha$-enlargement of $J_{n+1}$ remains in the target chart, and the coordinate convex hull of $\zeta_{n+1}(c_{n+1})$ with that enlargement lies in the target chart. In a boundary target chart, Lemma 9.1(iii) gives normal coordinate $\geq 2\alpha$ on this convex hull, a property preserved under any further convex combination. Property (iv) supplies a uniform-in-$n$ margin $\mu_*$ of the coordinate $4\alpha$-enlargement of $\chi_a(J_{n+1})$ from $\mathbb{R}^d \setminus \chi_a(U_a)$. By continuity of $f_n$ on the compact set $\overline{D_{n,1} \cup \cdots \cup D_{n,m}}$, shrink the $D_{n,i}$'s again if necessary, preserving separation, window containment, branch cubes and copy cores, so that

$$\operatorname{dist}_\infty\big(\zeta_{n+1}(f_n(D_{n,i})),\, \mathbb{R}^d \setminus \zeta_{n+1}(U_a)\big) \geq \mu_*/2.$$

Then every coordinate convex combination of a point of $\zeta_{n+1}(f_n(D_{n,i}))$ with a point of $\zeta_{n+1}(J_{n+1}^+(\theta))$ remains in the target chart, since both endpoints sit inside the coordinate $\mu_*/2$-neighborhood of the $4\alpha$-enlargement. In boundary charts the convex combination remains in $\mathbb{H}^d$ by convexity and the $2\alpha$ normal-coordinate margin. The shrinking constants depend only on $\alpha$, the atlas, $f_n$ and $W_n$, not on the absolute time index $n$ through any $n$-dependent constant.

On each $D_{n,i}$, define a model map $q_i$ in target coordinates exactly as in the cubical proof. If

$$\xi_n(B_{n,i}) = \prod_{\ell=1}^{d} [r_\ell, s_\ell]$$

and

$$\zeta_{n+1}(J_{n+1}) = \prod_{\ell=1}^{d} [A_{n+1,\ell}, B_{n+1,\ell}],$$

then $\zeta_{n+1}(q_i(x))$ is the clipped affine function of the source coordinates $\xi_n(x)$ taking the value $A_{n+1,\ell} - 2\theta$ on the lower $\ell$-face of $B_{n,i}$ and $B_{n+1,\ell} + 2\theta$ on the upper $\ell$-face. Thus $q_i(D_{n,i}) \subset J_{n+1}^+(\theta)$.

Write the source-coordinate cubes as

$$\xi_n(D_{n,i}) = \prod_{\ell=1}^{d} [u_\ell, v_\ell], \qquad \xi_n(C_{n,i}) = \prod_{\ell=1}^{d} [a_\ell, b_\ell].$$

Define $\eta_i : D_{n,i} \to D_{n,i}$ in source coordinates by

$$\big(\xi_n(\eta_i(x))\big)_\ell = u_\ell + (v_\ell - u_\ell)\,\operatorname{clip}_{[0,1]}\left(\frac{\big(\xi_n(x)\big)_\ell - a_\ell}{b_\ell - a_\ell}\right).$$

Then $\eta_i(C_{n,i}) = D_{n,i}$. Put $r_i = f_n \circ \eta_i$. Choose continuous bump functions $\varphi_i, \psi_i : D_{n,i} \to [0,1]$ such that $\varphi_i = 1$ on $B_{n,i}$, $\psi_i = 1$ on $C_{n,i}$, both vanish on $\partial D_{n,i}$, and $\varphi_i + \psi_i \leq 1$.

For $x \in D_{n,i}$, set

$$g_n(x) = \zeta_{n+1}^{-1}\Big((1 - \varphi_i(x) - \psi_i(x))\zeta_{n+1}(f_n(x)) + \varphi_i(x)\zeta_{n+1}(q_i(x)) + \psi_i(x)\zeta_{n+1}(r_i(x))\Big),$$

and put $g_n = f_n$ outside $\bigcup_i D_{n,i}$. The coordinate values lie in one convex target coordinate set; in boundary charts the convex set lies in $\mathbb{H}^d$. Since $\varphi_i$ and $\psi_i$ vanish on $\partial D_{n,i}$, the pieces glue continuously.

The coordinate displacement $\|\zeta_{n+1}(g_n(x)) - \zeta_{n+1}(f_n(x))\|_\infty$ is bounded above by the coordinate diameter of $\zeta_{n+1}(f_n(D_{n,i})) \cup \zeta_{n+1}(J_{n+1}^+(\theta))$, which is at most $L(B\alpha + (1+8\theta/\alpha)\alpha) \leq L(B+12)\alpha = C^*\alpha$, since $\theta = \alpha/20$. Both $q_i(x)$ and $r_i(x) = f_n(\eta_i(x))$ lie in this set; $C^*$ is independent of $n$ and $y$. Hence $D(f,g) \leq C^*\alpha < \eta$ by the choice of $\alpha$ at the start of the proof. Since $\varphi_i = 1$ on each branch cube, the strict coordinate face inequalities hold with coordinate margin $2\theta = \alpha/10$, uniformly in $n$ and $i$. The controlled choice of target boxes gives a uniform positive chart margin $\mu_* = \mu_{\mathrm{atlas}}(\alpha)$ and a target-chart $\mu_*/2$-neighborhood around every $g_n(B_{n,i})$, both independent of $n$. Hence $g \in \mathcal{U}_M(W_\bullet)$, and the strict manifold scheme has constants $(\alpha, \theta, \mu_*/2)$ which do *not* depend on the absolute time index $n$, even though the construction is carried out for all $n \in \mathbb{N}$ simultaneously.

Assume now that $f \in \mathcal{F}_s(M)$. Fix $n$. On $M \setminus \bigcup_i \operatorname{int} D_{n,i}$, the map $g_n$ agrees with $f_n$, and on each copy core,

$$g_n(C_{n,i}) = r_i(C_{n,i}) = f_n(\eta_i(C_{n,i})) = f_n(D_{n,i}).$$

Therefore

$$\begin{aligned} g_n(M) &\supset f_n\Big(M \setminus \bigcup_i \operatorname{int} D_{n,i}\Big) \cup \bigcup_i f_n(D_{n,i}) \\ &= f_n(M) = M. \end{aligned}$$

Thus $g_n$ is surjective for every $n$, and $g \in \mathcal{F}_s(M)$. □

**Lemma 9.6** (Manifold upper entropy spectrum)**.** *For every $f \in \mathcal{F}(M)$ and every $r \geq 0$,*

$$H_{f,r}(\varepsilon) \leq d \log(1/\varepsilon) + O_M(1) \qquad (\varepsilon \downarrow 0).$$

*Proof.* A compact Riemannian $d$-manifold, with or without boundary, has covering numbers

$$N(M, \rho, \varepsilon) \leq C_M \varepsilon^{-d}$$

for all sufficiently small $\varepsilon > 0$. Applying the same product-cover argument as in Lemma 8.5 gives

$$s_{r,N}(f, \varepsilon) \leq (C_M \varepsilon^{-d})^N.$$

The conclusion follows after taking logarithms, dividing by $N$, and taking $\limsup$. □

**Proposition 9.7** (Consequences of a strict manifold scheme)**.** *Let $g \in \mathcal{U}_M(W_\bullet)$, and fix a strict manifold horseshoe scheme with scale $\alpha$. Then every tail $\operatorname{tail}^r g$ admits a full variable-alphabet shift semifactor in the sense of Definition 2.3, has a continuum-sized DC1-scrambled set visiting the windows, and satisfies*

$$H_{g,r}(\Delta_m^M(\alpha)) \geq \log m \qquad (r \geq 0,\ m \geq 2).$$

*Consequently*

$$H_{g,r}(\varepsilon) \geq d \log(1/\varepsilon) - O_g(1) \qquad (\varepsilon \downarrow 0),$$

*with the additive constant independent of $r$.*

*Proof.* By Lemma 9.4, applied with $h = g$, every branch cube covers the next coordinate cube:

$$g_n(B_{n,i}) \supset J_{n+1} \qquad (n \geq 1,\ 1 \leq i \leq M_n).$$

Thus the backward compactness proof of Lemma 8.1 applies word-for-word, with coordinate cubes on $M$ replacing cubical subsets of $Q^d$. For each tail $r$, one obtains compact sets

$$\mathcal{K}_r^M = \{x \in J_{r+1} : G_r^j(x) \in \bigcup_{i=1}^{M_{r+j+1}} B_{r+j+1,i} \text{ for every } j \geq 0\}$$

and continuous surjections $\Pi_r^M : \mathcal{K}_r^M \to \prod_{n\geq r+1}\{1,\dots,M_n\}$. The branch cubes at each fixed time are compact and mutually $\rho$-separated, so itineraries are unique and cylinder preimages are open in $\mathcal{K}_r^M$. Hence

$$\Pi_{r+1}^M \circ g_{r+1} = \sigma_r \circ \Pi_r^M \quad \text{on } \mathcal{K}_r^M,$$

which is the asserted tail symbolic semifactor.

The DC1 construction is also the same explicit binary coding as in Proposition 8.3. On common $C$-blocks the two coded orbits lie in the same branch cube; since $\operatorname{diam}_\rho(B_{n,i}) < 1/n$, this gives upper distribution function equal to 1 at every positive scale. If two binary parameters differ in coordinate $s$, then on the infinitely many $D_s$- blocks the two coded orbits lie in different branches, whose Riemannian distance is greater than $\Delta_2^M(\alpha)$. The block condition $(S_k - r)/L_k \to 0$ gives lower distribution function 0 at that scale. The coded points lie in $W_{r+j+1}$ at tail time $j$ by construction.

For entropy, fix $m \geq 2$ and use an $E_m$-block after the initial tail time $r$. Each word in $\{1,\dots,m\}^{L_k}$ is realized by a point in $J_{r+1}$. Two different words disagree at some time in the block, and the corresponding orbit points then lie in different branches at Riemannian distance greater than $\Delta_m^M(\alpha)$. Hence there are $m^{L_k}$ points that are $(r, S_k + L_k - r, \Delta_m^M(\alpha))$-separated. Taking the limsup along $E_m$-blocks gives the displayed lower bound. Since $\Delta_m^M(\alpha) = c_{\alpha,M} m^{-1/d}$, the conversion from block scales $m^{-1/d}$ to arbitrary small $\varepsilon$ is exactly the argument in Proposition 8.4, with a constant independent of $r$. □

*Proof of Main Theorem 3.5.* By Lemma 9.4, $\mathcal{U}_M(W_\bullet)$ is open; by Lemma 9.5, it is dense. The relative openness of $\mathcal{U}_M(W_\bullet) \cap \mathcal{F}_s(M)$ in $\mathcal{F}_s(M)$ follows from the openness in the full space, and the relative density follows from the surjectivity-preserving part of Lemma 9.5.

Let $g \in \mathcal{U}_M(W_\bullet)$. The factor, windowed DC1, infinite-entropy, and entropy lower-spectrum conclusions are Proposition 9.7. The universal upper estimate is Lemma 9.6. Combining the two estimates gives, uniformly in the tail $r$,

$$d\log(1/\varepsilon) - O_g(1) \leq H_{g,r}(\varepsilon) \leq d\log(1/\varepsilon) + O_M(1).$$

Dividing by $\log(1/\varepsilon)$ and letting $\varepsilon \downarrow 0$ yields lower and upper metric mean dimension equal to $d$ for every tail. These conclusions apply in particular to the relative surjective open dense subset. □

## 10 Finite-support Hilbert boxes

We now turn to $Q^\infty$. A *finite-support Hilbert box* is a product

$$B = \prod_{j=1}^{\infty} I_j \subset Q^\infty$$

where each $I_j \subset [0,1]$ is a nondegenerate closed interval and $I_j = [0,1]$ for all but finitely many $j$. Such a box is compact and has nonempty interior in the Hilbert-cube topology. Its diameter satisfies

$$\operatorname{diam}_\rho(B) \leq \sum_{j=1}^{\infty} 2^{-j}|I_j|.$$

**Remark 10.1** (Why finite support is the right topology)**.** *The Hilbert metric induces the product topology on* $[0,1]^{\mathbb{N}}$*. A product box has nonempty interior exactly when it restricts only finitely* *many coordinates and each restricted coordinate interval has nonempty relative interior. Thus finite-support boxes are the natural local objects for dense-open window requirements. The general background on Hilbert cubes and Hilbert-cube manifolds is classical; see [13, 55, 56].*

For $K \geq 1$ and $0 < \alpha < 1$, an $(\alpha, K)$-box is a finite-support box of the form

$$J = \prod_{j=1}^{K} [a_j, a_j + \alpha] \times [0,1]^{\{K+1, K+2, \ldots\}}.$$

It has diameter at most $\alpha + 2^{-K}$.

**Lemma 10.2** (Small finite-support boxes form a local base)**.** *Let $U \subset Q^\infty$ be open, let $x \in U$, and let $\gamma > 0$. Then there is a finite-support Hilbert box $B$ such that*

$$x \in \operatorname{int} B \subset B \subset U, \qquad \operatorname{diam}_\rho(B) < \gamma.$$

*If finitely many points are separated in one of the first $q$ coordinates, the boxes around them may be chosen so small that this separation is preserved up to any prescribed positive loss.*

*Proof.* Since the metric $\rho$ induces the product topology, there are $R \geq 1$ and relative open intervals $V_1, \ldots, V_R \subset [0,1]$, with $x_j \in V_j$, such that

$$\prod_{j=1}^{R} V_j \times [0,1]^{\{R+1, R+2, \ldots\}} \subset U.$$

Choose $K \geq R$ so large that $2^{-K} < \gamma/2$. For $1 \leq j \leq K$, choose nondegenerate closed intervals $I_j \subset V_j$ for $j \leq R$ and $I_j \subset [0,1]$ for $R < j \leq K$, each containing $x_j$ in its relative interior and with

$$\sum_{j=1}^{K} 2^{-j} |I_j| < \gamma/2.$$

Set $I_j = [0,1]$ for $j > K$. Then $B = \prod I_j$ has the required properties. The final assertion follows by choosing the first $q$ intervals small enough around the prescribed points. □

**Lemma 10.3** (Windowed finite-coordinate selection in $Q^\infty$)**.** *Fix $0 < \alpha < 1/4$ and $q \geq 1$. There exists a constant $c_{\alpha,q} > 0$ with the following property. Let $J \subset Q^\infty$ be an $(\alpha, K)$-box with $K \geq q$, let $W \subset Q^\infty$ be dense open, let $f \in C(Q^\infty, Q^\infty)$, and let $m \geq 2$. Then there are points*

$$p_1, \ldots, p_m \in \operatorname{int}(J) \cap W$$

*such that*

$$\max_{1 \leq \ell \leq q} |(p_i)_\ell - (p_j)_\ell| > 4 c_{\alpha,q} m^{-1/q} \qquad (i \neq j),$$

*and*

$$\operatorname{diam}_\rho \{f(p_1), \ldots, f(p_m)\} < \alpha/16.$$

*Moreover, the points may be thickened to pairwise disjoint finite-support boxes inside* $\operatorname{int}(J) \cap W$ *while preserving the separation with $4c_{\alpha,q}$ replaced by $c_{\alpha,q}$.*

*Proof.* Write

$$J = \prod_{j=1}^{K} [u_j, u_j + \alpha] \times [0, 1]^{\{K+1, K+2, \ldots\}}.$$

Define a finite-dimensional slice $\iota : P \to Q^\infty$ as follows. Let

$$P = \prod_{\ell=1}^{q} [u_\ell + \alpha/4, u_\ell + 3\alpha/4].$$

For $z = (z_1, \ldots, z_q) \in P$, set $(\iota z)_\ell = z_\ell$ for $1 \le \ell \le q$, set $(\iota z)_j = u_j + \alpha/2$ for $q < j \le K$, and set $(\iota z)_j = 1/2$ for $j > K$. The image of $\iota$ lies in $\operatorname{int} J$.

By compactness, cover $Q^\infty$ by finitely many Borel sets $R_1, \ldots, R_N$, each of $\rho$-diameter less than $\alpha/64$. The measurable sets

$$A_s = \{z \in P : f(\iota z) \in R_s\}, \qquad 1 \le s \le N,$$

cover $P$. Hence for some $s$,

$$\mathrm{Leb}_q(A_s) \ge \frac{(\alpha/2)^q}{N}.$$

By Lemma 5.1, $A_s$ contains points $z_1, \ldots, z_m$ pairwise more than

$$a_q \left( \frac{(\alpha/2)^q}{N} \right)^{1/q} m^{-1/q}$$

apart in sup norm. Choose

$$c_{\alpha,q} \le \frac{a_q}{8} \left( \frac{(\alpha/2)^q}{N} \right)^{1/q}.$$

Then the first $q$ coordinates of $\iota z_i$ and $\iota z_j$ are separated by more than $8 c_{\alpha,q} m^{-1/q}$, and all $f(\iota z_i)$ lie in one $R_s$, so their image diameter is less than $\alpha/64$.

By continuity of $f$, choose pairwise disjoint open neighborhoods $U_i \subset \operatorname{int} J$ of $\iota z_i$ such that the first $q$-coordinate separation between different $U_i$'s is still larger than $4 c_{\alpha,q} m^{-1/q}$, and

$$\rho(f(u), f(\iota z_i)) < \alpha/64 \qquad (u \in U_i).$$

Since $W$ is dense open, choose $p_i \in U_i \cap W$. Then

$$\mathrm{diam}_\rho \{f(p_1), \ldots, f(p_m)\} < 3\alpha/64 < \alpha/16.$$

Finally, $\operatorname{int} J \cap W \cap U_i$ is an open neighborhood of $p_i$. By Lemma 10.2, choose small finite-support boxes around the $p_i$'s inside these neighborhoods, pairwise disjoint and small enough in the first $q$ coordinates to preserve the stated separation with constant $c_{\alpha,q}$. □

## 11 Exact Hilbert-box branch implantation

**Lemma 11.1** (Coordinatewise maps onto Hilbert boxes)**.** *Let*

$$B = \prod_{j=1}^{\infty} [a_j, b_j]$$

*be a finite-support Hilbert box, with every interval nondegenerate, and let*

$$J' = \prod_{j=1}^{\infty}[a_j', b_j']$$

*be another finite-support Hilbert box. Define*

$$\lambda_j(t) = \mathrm{clip}_{[0,1]}\left(\frac{t - a_j}{b_j - a_j}\right), \qquad q_j(x) = a_j' + (b_j' - a_j')\lambda_j(x_j),$$

*and* $q(x) = (q_1(x), q_2(x), \ldots)$. *Then* $q : Q^\infty \to Q^\infty$ *is continuous,* $q(Q^\infty) \subset J'$, *and* $q(B) = J'$.

*Proof.* Each coordinate map $q_j$ is continuous and takes values in $[a_j', b_j']$, so $q$ is continuous for the product topology, equivalently for the Hilbert metric $\rho$. The inclusion $q(Q^\infty) \subset J'$ is immediate. For the reverse inclusion, let $y \in J'$. For each $j$, choose $x_j \in [a_j, b_j]$ such that

$$a_j' + (b_j' - a_j')\lambda_j(x_j) = y_j,$$

which is possible because the affine coordinate map sends $[a_j, b_j]$ onto $[a_j', b_j']$. Then $x = (x_j)_j \in B$ and $q(x) = y$. □

**Lemma 11.2** (Finite Urysohn gluing in the Hilbert cube)**.** *Let* $B_1, \ldots, B_m$ *be compact subsets of pairwise disjoint open sets* $O_1, \ldots, O_m \subset Q^\infty$. *Let* $f, q_i \in C(Q^\infty, Q^\infty)$. *Then there is* $g \in C(Q^\infty, Q^\infty)$ *such that* $g = q_i$ *on* $B_i$, $g = f$ *outside* $\bigcup_i O_i$, *and, if*

$$\sup_{x \in O_i} \rho(q_i(x), f(x)) \le \eta_i,$$

*then* $\sup_x \rho(g(x), f(x)) \le \max_i \eta_i$.

*Proof.* The Hilbert cube is compact metric and therefore normal. Choose continuous functions $\varphi_i : Q^\infty \to [0,1]$ with $\varphi_i = 1$ on $B_i$ and $\varphi_i = 0$ outside $O_i$. Since the $O_i$'s are pairwise disjoint, $\sum_i \varphi_i \le 1$. Define the coordinatewise convex combination

$$g(x) = \left(1 - \sum_i \varphi_i(x)\right) f(x) + \sum_i \varphi_i(x) q_i(x).$$

The product $[0,1]^{\mathbb{N}}$ is convex, so $g(x) \in Q^\infty$, and coordinatewise continuity gives continuity in the Hilbert metric. The identities on the $B_i$'s and outside $\bigcup_i O_i$ follow from the choice of the bump functions. The distance estimate follows from convexity of the metric $\rho$, which is the weighted $\ell^1$-sum of coordinate distances. □

**Lemma 11.3** (Exact Hilbert-box implantation)**.** *Let* $0 < \alpha < 1/4$, *let* $J \subset Q^\infty$ *be an* $(\alpha, K)$*-box, let* $W \subset Q^\infty$ *be dense open, let* $f \in C(Q^\infty, Q^\infty)$, *let* $q \ge 1$, *let* $m \ge 2$, *and let* $\gamma > 0$. *Assume* $K \ge q$ *and* $K' \ge q$. *The parameter* $K'$ *is otherwise free; no additional "sufficiently large" hypothesis on* $K'$ *is used in the proof, and the constants* $c_{\alpha,q}$ *and the conclusion (11.1) depend on* $K'$ *only through the explicit tail term* $2^{-K'}$. *Under* $K \ge q$ *and* $K' \ge q$ *there are:*

(i) *pairwise disjoint finite-support Hilbert boxes*

$$B_1, \ldots, B_m \subset \mathrm{int}(J) \cap W$$

*with* $\mathrm{diam}_\rho(B_i) < \gamma$;

(ii) *an* $(\alpha, K')$*-box* $J'$*;*

(iii) *a map* $g \in C(Q^\infty, Q^\infty)$*;*

*such that*

$$\sup_{x\in Q^\infty} \rho(g(x), f(x)) < 2\alpha + 2^{-K'}, \tag{11.1}$$

$$g(B_i) = J' \qquad (1 \le i \le m),$$

*and*

$$\operatorname{dist}_q(B_i, B_j) := \inf_{x\in B_i,\ y\in B_j} \max_{1\le \ell\le q} |x_\ell - y_\ell| > c_{\alpha,q} m^{-1/q}$$

*for* $i \neq j$*.*

*Proof.* By Lemma 10.3, choose points $p_1, \dots, p_m \in \operatorname{int}(J) \cap W$ whose first $q$ coordinates are pairwise more than $4c_{\alpha,q}m^{-1/q}$ apart and whose images under $f$ have $\rho$-diameter less than $\alpha/16$. By continuity choose pairwise disjoint open neighborhoods $O_i \subset \operatorname{int}(J) \cap W$ of $p_i$ such that the first $q$-coordinate separation between different $O_i$'s is larger than $c_{\alpha,q}m^{-1/q}$, and

$$\rho(f(x), f(p_i)) < \alpha/32 \qquad (x \in O_i).$$

Then

$$\operatorname{diam}_\rho f(O_1 \cup \dots \cup O_m) < \alpha/16 + 2(\alpha/32) = \alpha/8.$$

Using Lemma 10.2, choose pairwise disjoint finite-support boxes $B_i \subset O_i$ with $p_i \in \operatorname{int} B_i$ and $\operatorname{diam}_\rho(B_i) < \gamma$. The stated $q$-coordinate separation follows from the choice of the $O_i$'s.

Let $c = f(p_1)$. For the prescribed sufficiently large $K' \ge q$, define

$$J' = \prod_{j=1}^{K'} [s_j, s_j + \alpha] \times [0, 1]^{\{K'+1, K'+2, \dots\}}$$

by

$$s_j = \min\{\max\{c_j - \alpha/2, 0\}, 1 - \alpha\}, \qquad 1 \le j \le K'.$$

Every point of $J'$ is within $\rho$-distance at most $\alpha + 2^{-K'}$ of $c$: the first $K'$ coordinates contribute at most $\alpha$, and the tail contributes at most $2^{-K'}$.

For each branch box $B_i$, let $q_i : Q^\infty \to Q^\infty$ be the coordinatewise map of Lemma 11.1 from $B_i$ onto $J'$. Then $q_i$ is continuous and $q_i(B_i) = J'$. If $x \in O_i$, then $\rho(f(x), c) < \alpha/8$, while $q_i(x) \in J'$ gives $\rho(q_i(x), c) \le \alpha + 2^{-K'}$. Therefore

$$\rho(q_i(x), f(x)) < 2\alpha + 2^{-K'}.$$

Applying Lemma 11.2 to the boxes $B_i$, neighborhoods $O_i$, and maps $q_i$, gives a continuous map $g$ with $g = q_i$ on $B_i$, $g(B_i) = J'$, and

$$\sup_{x\in Q^\infty} \rho(g(x), f(x)) < 2\alpha + 2^{-K'}.$$

$\square$

## 12 A Hilbert-cube universal block schedule

For the Hilbert theorem use the countable label set

$$\Lambda_\infty = \{C_a : a \geq 1\} \cup \{D_s : s \geq 1\} \cup \{E_{q,m} : q \geq 1,\ m \geq 2\}.$$

Choose $\lambda_k \in \Lambda_\infty$ so that every label occurs infinitely often. Choose positive integers $L_k$ with $S_k/L_k \to 0$, where $S_k = L_1 + \cdots + L_{k-1}$, and put

$$T_k = \{S_k + 1, \ldots, S_k + L_k\}.$$

For $n \in T_k$, define

$$(M_n, q_n) = \begin{cases} (2,1), & \lambda_k = C_a, \\ (2,1), & \lambda_k = D_s, \\ (m,q), & \lambda_k = E_{q,m}. \end{cases}$$

Set

$$\gamma_n = \begin{cases} \min\{1/n, 1/a\}, & \lambda_{k(n)} = C_a, \\ 1/n, & \text{otherwise.} \end{cases}$$

As before, deleting finitely many initial times does not change the block density estimates.

**Definition 12.1** (Exact windowed Hilbert horseshoe). *A system $g = (g_n) \in \mathcal{F}(Q^\infty)$ has an exact $W_\bullet$-windowed Hilbert horseshoe if there are $0 < \alpha < 1/4$, integers $K_n \geq q_n$, $(\alpha, K_n)$-boxes $J_n$, and finite-support boxes*

$$B_{n,1}, \ldots, B_{n,M_n} \subset \operatorname{int}(J_n) \cap W_n$$

*such that:*

(1) $\operatorname{diam}_\rho(B_{n,i}) < \gamma_n$ *for all* $n, i$.

(2) *For* $i \neq j$,

$$\operatorname{dist}_{q_n}(B_{n,i}, B_{n,j}) > c_{\alpha,q_n} M_n^{-1/q_n}.$$

(3) *For every* $n$ *and* $i$,

$$g_n(B_{n,i}) = J_{n+1}.$$

**Proposition 12.2** (Dense exact Hilbert implantation). *For every sequence of dense open windows $W_n \subset Q^\infty$, the set of systems in $\mathcal{F}(Q^\infty)$ admitting an exact $W_\bullet$-windowed Hilbert horseshoe is dense.*

*Proof.* Let $f = (f_n) \in \mathcal{F}(Q^\infty)$ and let $\eta > 0$. Choose

$$0 < \alpha < \min\{1/4, \eta/4\}.$$

Choose $K_1 \geq q_1$ and an arbitrary $(\alpha, K_1)$-box $J_1$; the inductive hypothesis is that $J_n$ is an $(\alpha, K_n)$-box with $K_n \geq q_n$. Suppose $J_n$ has been chosen. Set

$$K_{n+1} \ = \ \max\Big\{q_n,\ q_{n+1},\ \big\lceil \log_2(1/\alpha) + 1 \big\rceil\Big\}, \tag{12.1}$$

so that the three explicit requirements

(K1) $K_{n+1} \geq q_n$, required to apply Lemma 11.3 with $K' = K_{n+1}$ and parameter $q = q_n$;

(K2) $K_{n+1} \geq q_{n+1}$, required so that the produced $(\alpha, K_{n+1})$-box $J_{n+1}$ is admissible as the input box at step $n+1$, where the lemma will be applied with parameter $q = q_{n+1}$; and

(K3) $2^{-K_{n+1}} < \alpha$, required for the perturbation size estimate below

are all met. These are the only constraints on $K_{n+1}$; by the closing clause of Lemma 11.3, no implicit "sufficiently large" hypothesis on $K' = K_{n+1}$ is in force. Apply Lemma 11.3 to $f_n$, $J_n$, $W_n$, $q_n$, $M_n$, $\gamma_n$, and $K' = K_{n+1}$; the lemma's hypotheses $K_n \geq q_n$ and $K_{n+1} \geq q_n$ both hold by the inductive hypothesis and (K1). We obtain branch boxes $B_{n,i}$, a next box $J_{n+1}$ which is an $(\alpha, K_{n+1})$-box, and a map $g_n$ with

$$\sup_x \rho(g_n(x), f_n(x)) < 2\alpha + 2^{-K_{n+1}} < 3\alpha < \eta$$

by (11.1) and (K3). Because of (K2), the inductive hypothesis is preserved: $J_{n+1}$ is an $(\alpha, K_{n+1})$-box with $K_{n+1} \geq q_{n+1}$. Doing this independently for every $n$ produces $g = (g_n)$ with $D(f, g) < \eta$, and by construction $g$ has an exact windowed Hilbert horseshoe. $\square$

## 13 Hilbert-cube coding, DC1, and entropy

Fix a system $g$ with an exact Hilbert horseshoe. For $r \geq 0$, put

$$G_r^0 = \mathrm{id}_{Q^\infty}, \qquad G_r^j = g_{r+j} \circ \cdots \circ g_{r+1} \quad (j \geq 1).$$

Let

$$\Sigma_r^\infty = \prod_{n \geq r+1} \{1, \ldots, M_n\}.$$

**Lemma 13.1** (Hilbert tail itinerary coding)**.** *Fix $r \geq 0$. For every itinerary*

$$\omega = (\omega_{r+1}, \omega_{r+2}, \ldots), \qquad \omega_n \in \{1, \ldots, M_n\},$$

*there exists $x_\omega \in J_{r+1}$ such that*

$$G_r^j(x_\omega) \in B_{r+j+1, \omega_{r+j+1}} \subset W_{r+j+1} \qquad (j \geq 0).$$

*If two itineraries differ at some time, no point realizes both.*

*Proof.* The proof is the same backward compactness argument as Lemma 8.1, using equality $g_n(B_{n,i}) = J_{n+1}$. For finite length $N$, choose $y_N \in B_{r+N, \omega_{r+N}}$ and pull it back successively through the surjective restrictions

$$g_{r+j}(B_{r+j, \omega_{r+j}}) = J_{r+j+1}.$$

This gives a nonempty compact set of points realizing the first $N$ symbols. The finite realizing sets are nested, so their intersection is nonempty. Two different itineraries cannot be realized by one point because branch boxes at the time of first disagreement are disjoint. $\square$

**Proposition 13.2** (Hilbert full shift semifactor)**.** *For every $r \geq 0$, define*

$$\mathcal{K}_r^\infty = \{x \in J_{r+1} : G_r^j(x) \in \bigcup_{i=1}^{M_{r+j+1}} B_{r+j+1,i} \text{ for every } j \geq 0\}.$$

*Then $\mathcal{K}_r^\infty$ is a nonempty compact set, $g_{r+1}(\mathcal{K}_r^\infty) \subset \mathcal{K}_{r+1}^\infty$, and there is a continuous surjection*

$$\Pi_r^\infty : \mathcal{K}_r^\infty \to \Sigma_r^\infty$$

*intertwining the non-autonomous dynamics with the left shift.*

*Proof.* Compactness and forward invariance are immediate from the definition. Existence of every itinerary is Lemma 13.1. Since at each time the branch boxes are compact and pairwise separated in at least one finite coordinate, they have positive $\rho$-distance. Therefore the itinerary of each point in $\mathcal{K}_r^\infty$ is unique and depends continuously on the point by the same cylinder-set argument used in Proposition 8.2. Surjectivity and the intertwining identity follow from the construction. $\square$

**Proposition 13.3** (Hilbert all-tail windowed DC1)**.** *For every $r \geq 0$, the tail $\mathrm{tail}^r g$ has a DC1-scrambled set of cardinality $\mathfrak{c}$, and the scrambled set may be chosen so that*

$$G_r^j(x) \in W_{r+j+1} \qquad (j \geq 0).$$

*Proof.* For each binary sequence $\beta = (\beta_1, \beta_2, \ldots)$, define an itinerary for times $n \geq r+1$ as follows. On a $C_a$-block set $\omega_n^\beta = 1$. On a $D_s$-block set $\omega_n^\beta = 1 + \beta_s$. On every entropy block $E_{q,m}$, set $\omega_n^\beta = 1$. Choose a realizing point $x_\beta^{(r)}$ from Lemma 13.1 and set

$$\mathcal{S}_r^\infty = \{x_\beta^{(r)} : \beta \in \{0,1\}^{\mathbb{N}}\}.$$

The window condition follows from the coding lemma. If $\beta \neq \gamma$, choose $s$ with $\beta_s \neq \gamma_s$. Since $D_s$ occurs infinitely often, the corresponding itineraries differ after time $r$; hence the realizing points are distinct. Thus $\mathrm{card}(\mathcal{S}_r^\infty) = \mathfrak{c}$.

Let $x = x_\beta^{(r)}$, $y = x_\gamma^{(r)}$, and let $t > 0$. Choose $a$ with $1/a < t$. Along every sufficiently late $C_a$-block, both itineraries choose branch 1, and the branch diameter is at most $\gamma_n \leq 1/a < t$. Since these blocks have asymptotic density one along their endpoints, $(\psi_{xy}^r)^*(t) = 1$.

For separation, choose $s$ with $\beta_s \neq \gamma_s$. On every sufficiently late $D_s$-block the two itineraries choose branches 1 and 2, which are separated in the first coordinate by more than $c_{\alpha,1}2^{-1}$. Consequently their Hilbert distance is more than

$$\delta_0 = 2^{-1}c_{\alpha,1}2^{-1} = c_{\alpha,1}/4.$$

Throughout these $D_s$-blocks the two orbits are not $\delta_0$-close, and again the block-density estimate gives $\psi_{xy}^r(\delta_0) = 0$. Hence the pair is DC1-scrambled. $\square$

**Proposition 13.4** (Hilbert finite-coordinate entropy lower bounds)**.** *Let $g$ admit an exact Hilbert horseshoe with parameter $\alpha$. Then for every tail $r \geq 0$, every finite $q \geq 1$, and every $m \geq 2$,*

$$H_{g,r}\big(2^{-q}c_{\alpha,q}m^{-1/q}\big) \geq \log m.$$

*Consequently, for every $q \geq 1$ there is a constant $C_{g,q}$, independent of $r$, such that*

$$H_{g,r}(\varepsilon) \geq q\log(1/\varepsilon) - C_{g,q} \qquad (r \geq 0,\ \varepsilon \downarrow 0).$$

*Thus every tail has infinite topological entropy and infinite lower and upper metric mean dimension.*

*Proof.* Fix $r, q, m$. The label $E_{q,m}$ occurs infinitely often. Let $T_k = \{S_k+1, \ldots, S_k+L_k\}$ be such a block with $S_k + 1 \geq r+1$. For each word

$$u = (u_{S_k+1}, \ldots, u_{S_k+L_k}) \in \{1, \ldots, m\}^{L_k},$$

use the finite form of Lemma 13.1 to choose a point $x_u \in J_{r+1}$ whose itinerary uses $u_n$ on this block and symbol 1 elsewhere before the end of the block. If $u \neq v$, choose $n \in T_k$ with $u_n \neq v_n$. At tail time $j = n - r - 1$, the two orbit points lie in two distinct branch boxes of an $E_{q,m}$-block. Their first $q$ coordinates are separated in sup norm by more than $c_{\alpha,q}m^{-1/q}$. Therefore their Hilbert distance is more than

$$\varepsilon_{q,m} = 2^{-q}c_{\alpha,q}m^{-1/q}.$$

The set of all $x_u$ is $(r, S_k + L_k - r, \varepsilon_{q,m})$-separated and has cardinality $m^{L_k}$. Taking logarithms, dividing by $S_k + L_k - r$, and passing to the limsup along $E_{q,m}$-blocks gives

$$H_{g,r}(\varepsilon_{q,m}) \geq \log m.$$

Put $A_q = 2^{-q}c_{\alpha,q}$. If $A_q(m+1)^{-1/q} < \varepsilon \leq A_q m^{-1/q}$, then monotonicity gives

$$H_{g,r}(\varepsilon) \geq H_{g,r}(A_q m^{-1/q}) \geq \log m.$$

Since $m \geq (A_q/\varepsilon)^q - 1$, for all sufficiently small $\varepsilon$,

$$H_{g,r}(\varepsilon) \geq q \log(1/\varepsilon) - C_{g,q},$$

where $C_{g,q}$ depends only on $A_q$, hence only on $g$'s horseshoe parameter $\alpha$ and on $q$, and not on $r$. Because $q$ is arbitrary, $\underline{\mathrm{mdim}}_{\mathrm{M}}(\mathrm{tail}^r\, g) \geq q$ for every $q$, hence both lower and upper metric mean dimensions are infinite. Infinite entropy follows from the same estimates. □

*Proof of Main Theorem 3.6.* By Proposition 12.2, systems admitting exact windowed Hilbert horseshoes are dense. For every such system, Proposition 13.2 gives the full shift semifactors in every tail, Proposition 13.3 gives all-tail windowed DC1 chaos, and Proposition 13.4 gives the entropy lower bounds, infinite topological entropy, and infinite lower and upper metric mean dimension. □

## 14 Sharpness and scope

The interval results are now part of the main paper rather than a separate input. The full interval theorem is exactly the cubical theorem with $d = 1$, and the surjective interval theorem is the $d = 1$ case of the surjectivity-preserving cubical implantation. Thus the earlier non-tail interval conclusions are recovered by taking $r = 0$, while the present version also proves the same symbolic, DC1, entropy, and metric-mean-dimension conclusions for every forward tail and proves the relative surjective statement in every finite cubical dimension.

In finite dimension the exponent $d$ is optimal for the fixed metric geometry: Lemma 8.5 and Lemma 9.6 bound the entropy spectrum of every system from above by $d \log(1/\varepsilon) + O(1)$, while the strict horseshoe construction attains the matching lower bound. In the Hilbert cube there is no finite metric dimension to saturate. The construction instead realizes every finite-dimensional entropy exponent $q$ along suitable blocks and hence forces infinite lower and upper metric mean dimension in every tail.

The finite-dimensional conclusions are open dense because strict face margins are protected by a single positive uniform perturbation radius. The Hilbert-cube conclusion is dense but is not asserted to be open: exact covering in arbitrarily high coordinates is not stable under the Hilbert metric. A residual Hilbert-cube theorem with coherent all-tail symbolic codings would require an additional Baire-fusion mechanism beyond the exact finite-support implantations proved here.

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## Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the author used generative AI tools to accelerate drafting and revision. All mathematical claims, proofs, and bibliographic information were subsequently reviewed and edited by the author, who takes full responsibility for the content of the manuscript.

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